DunenScope

User manual

Version 0.8.0 — November 2026

DunenScope is a Windows tool for diagnosing and commissioning Dunen / ELYX motor controllers. It uses the same protocol as DunenConfiger and complements it. Since version 0.8, it also reads other devices: controllers from other brands, industrial drives, batteries and chargers, machinery (chapter 16).

1. Overview

DunenScope is used to read, understand and fine-tune a Dunen / ELYX controller from a Windows PC, during machine commissioning or after-sales service. It is intended for machine builders who integrate these controllers and for their technicians.

DunenConfiger remains the reference configuration tool. DunenScope reads the same parameter file, with the same row numbers, and its xlsx exports can be re-imported into DunenConfiger. It adds monitoring, diagnostic and tuning functions:

Area	What DunenScope adds
Plots	Six synchronised plots, a custom plot with up to 4 variables of your choice, CSV recording
Dashboard	Speed, torque, voltage, current, throttle, power, temperatures, state and faults on one screen
Diagnostics	"Why is it capped?": which limit is active (torque, battery current, speed, voltage) and for how long; battery, wired inputs and temperatures; last fault analysis
Setup wizard	Step-by-step assistant: motor, controller and battery, command and pedal calibration, machine, writing, encoder zero, direction of rotation, PI tuning
Simple settings	The whole controller by category, in the machine's units (s, %, km/h, L/min, m/min), mouse-editable curves, assistants
Comparison	Differences against a parameter file; end-of-line test with report and production log
Support	Complete support package in one click (.zip)
Speed PI tuning	Measurement assistant on the machine, manual step tuning and automatic tuning
Profiles	Application profiles (machine + CAN adapter) that can be exported to another PC

Area	What DunenScope adds
Other devices	Controllers compatible with VESC® Software, Kelly, Votol, Fardriver, Bafang, ODrive; CANopen® and Modbus drives; batteries and chargers; J1939 machinery; any CAN bus described by a DBC file
Overview	The device's key values, the other controllers on the bus, fault log and black box of the 20 s before each fault
Oscilloscope	4 channels, triggered by a threshold or by a fault, cursors, CSV export
CAN analyser	Listens to the bus without transmitting anything, identifies the frames, CSV trace
Commissioning report	PDF to hand over to the customer: controller, checks, measurements, settings, faults, signatures
Online library	Model profiles shared by other users (without their values)
Languages	Interface in 10 languages: French, English, Chinese, Spanish, German, Italian, Portuguese (Brazil), Russian, Japanese, Korean
CAN adapters	PEAK PCAN-USB, Kvaser, IXXAT, Vector, candleLight, SLCAN (CANable, DunenScope Link...), USB-CAN Analyzer

2. Installation

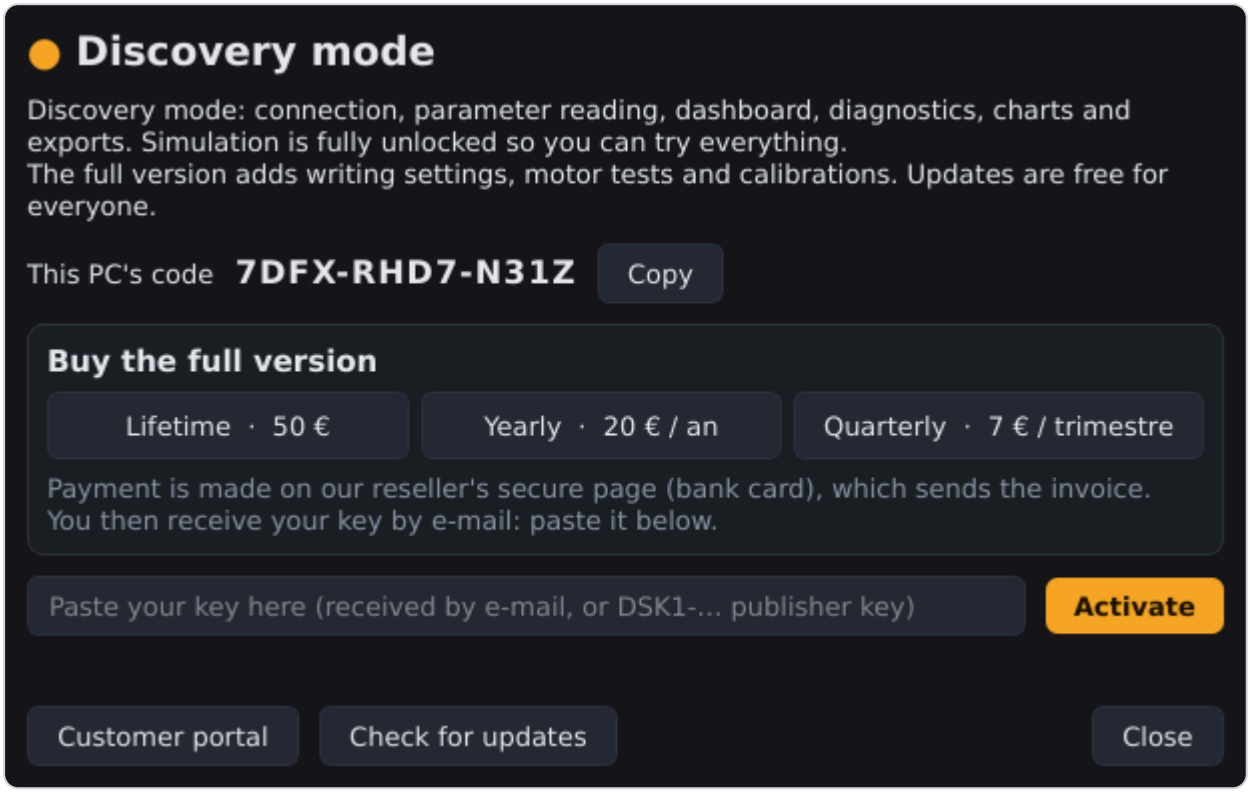
- A single file to run: **DunenScope_Setup_0.8.0.exe**. A setup wizard in 10 languages (language, licence agreement, folder, shortcuts) creates the shortcut and the manual in the Start menu; it can be uninstalled from Windows "Apps". The application is delivered compiled and ready to use.
- A new version installs over the old one, manually or automatically (section 3): settings, profiles and backups are kept.
- The interface is available in 10 languages (list at the top right): French, English, Chinese, Spanish, German, Italian, Portuguese (Brazil), Russian, Japanese, Korean. By default, DunenScope uses the Windows display language (first entry of the list, "(auto)"); choosing a specific language keeps it from then on.
- If the PEAK driver for the PCAN-USB adapter is not found, the installer offers to install it. DunenScope also offers it at connection time if it is missing.
- Kvaser, IXXAT, Vector: install the driver supplied by their manufacturer. candleLight, CANable, DunenScope Link: nothing to install. USB-CAN Analyzer: driver of its USB-serial chip (CH340 / CH341).
- Serial links (Modbus RTU, BMS, Kelly, Fardriver, Victron...): a USB-RS485 adapter or the manufacturer's cable, seen by Windows as a COM port.
- The python-can library (LGPL licence) is shipped separately, in the "bibliotheques" folder of the installation folder, and can be replaced as its licence allows. If it is missing, DunenScope says so: reinstall.
- Numbers: both point and comma are accepted in every field, whatever the Windows language.
- The window fits a laptop screen (1280 × 680).
- System requirements: Windows 10 or Windows 11, 64-bit.

3. Licence, purchase and updates

DunenScope installs and runs free of charge in **discovery mode**: connection, reading of parameters and of other devices, dashboard, overview, diagnostics, charts, oscilloscope, CAN analyser, exports, support package and commissioning report. In **simulation**, every feature is unlocked so you can try it. The **full version** adds everything that writes to a real controller (settings, motor tests, calibrations), including on the Other device page. Updates are free for everyone, with or without a key.

The button at the top right ("Licence: discovery" or "Licence: full") opens the Licence window:

Item	Purpose
This PC's code	Short PC identifier (12 characters), to give when asking for a key tied to this PC
Buy	Lifetime, Yearly or Quarterly: opens the secure payment page; the key arrives by e-mail
Activate	Paste the key received (purchased key, or DSK1-... publisher key), then "Activate"
Release this PC	Makes the key usable on another PC (internet required)
Customer portal	Invoices, subscription, activated PCs
Check for updates	Checks right away whether a new version has been published



Licence window (discovery mode)

- **One key = one PC.** A purchased key is activated once over the internet (a phone's hotspot is enough) and stays tied to that PC. On another PC it is refused until the first one has been released.

- **Without internet.** Lifetime purchase: DunenScope then works without internet. Subscription: an online check is needed at least every 45 days; DunenScope warns 10 days ahead.
- **Publisher keys (DSK1-...).** Supplied by the publisher (trials, partners): checked without internet, optionally limited to one PC and/or until a date.
- **End of licence.** Subscription stopped, key disabled or expired: DunenScope returns to discovery mode; backups, profiles and files are kept.
- **Automatic updates (free for everyone).** With internet, in discovery mode as well as in the full version, DunenScope checks at start-up whether a new version has been published and offers it ("Install now", "Later", "Skip this version"). The installer is downloaded and its signature checked (an altered file is refused), then it replaces the old version and restarts DunenScope. Windows asks for permission to install.

4. First connection

1. Key ON, adapter plugged into the controller's CAN connector.
2. Leave the top bar on **"Adapter: auto"** and **"Controller: auto"**, then click **"Connect"** (or the same button in the dashboard banner; "Try without hardware (simulation)" starts the simulation). DunenScope finds by itself the adapter that is plugged in (PEAK PCAN-USB, Kvaser, IXXAT, Vector, candleLight or CANable), the bus bit rate and the controller address (node). The search only reads: nothing is written to the controller.
3. The indicator next to the button changes to "Connected". The status bar at the bottom shows "Link OK" and the number of samples per second (in orange if the link becomes unstable), then the model, software and hardware versions and the serial number; the log shows what was found (for example "PCAN_USBBUS1, 250 kbit/s, node 1").
4. The model is recognised and its parameter file loaded automatically. The model can also be chosen in the "Controller" list.

The last choice of the adapter list, "Channel, bit rate, node...", sets the channel, bit rate and node by hand, which is only useful when several controllers share the same bus. The next connection first tries the settings that worked last time.

The bit rate is found by listening to the bus without transmitting anything. IXXAT, CANable and USB-CAN Analyzer cannot do that: choose their bit rate in "Channel, bit rate, node..." (a wrong bit rate would disturb the bus). For a USB-CAN Analyzer, also choose its COM port there.

DunenScope and DunenConfiger cannot use the adapter at the same time: close one before opening the other.

To try the application without hardware, select the "Simulation (no hardware)" adapter: a simulated controller feeds every page. The screenshots in this manual were taken in this mode.

Unknown controller or other device

If the connected controller is not yet known to DunenScope, it stays **read-only** so that nothing is damaged, and offers the "Other device" mode. If no Dunen answers but a CANopen® drive, a controller compatible with VESC® Software or a Kelly is talking on the bus, DunenScope offers to connect to it. For any other device, choose "Other device..." in the "Controller" list. See chapter 16.

Help, contact and error reports

The **Help** button in the top bar opens the "Help and contact" window: version, manual, software page and contact address. DunenScope is brand new: to report a problem, suggest an idea or ask a question, write directly in this window and click "Send". The technical information and the session log are attached (untick the box if needed, "Show what will be sent" shows everything); no personal data is sent. Without internet, the message is kept and sent at the next start.

If an unexpected error occurs, or if DunenScope closed abruptly, a window offers to send the error report. Nothing is sent without your consent; the "Always send error reports without asking" box skips the question next time.

5. Dashboard

The Dashboard continuously shows the main values read from the controller.



Dashboard

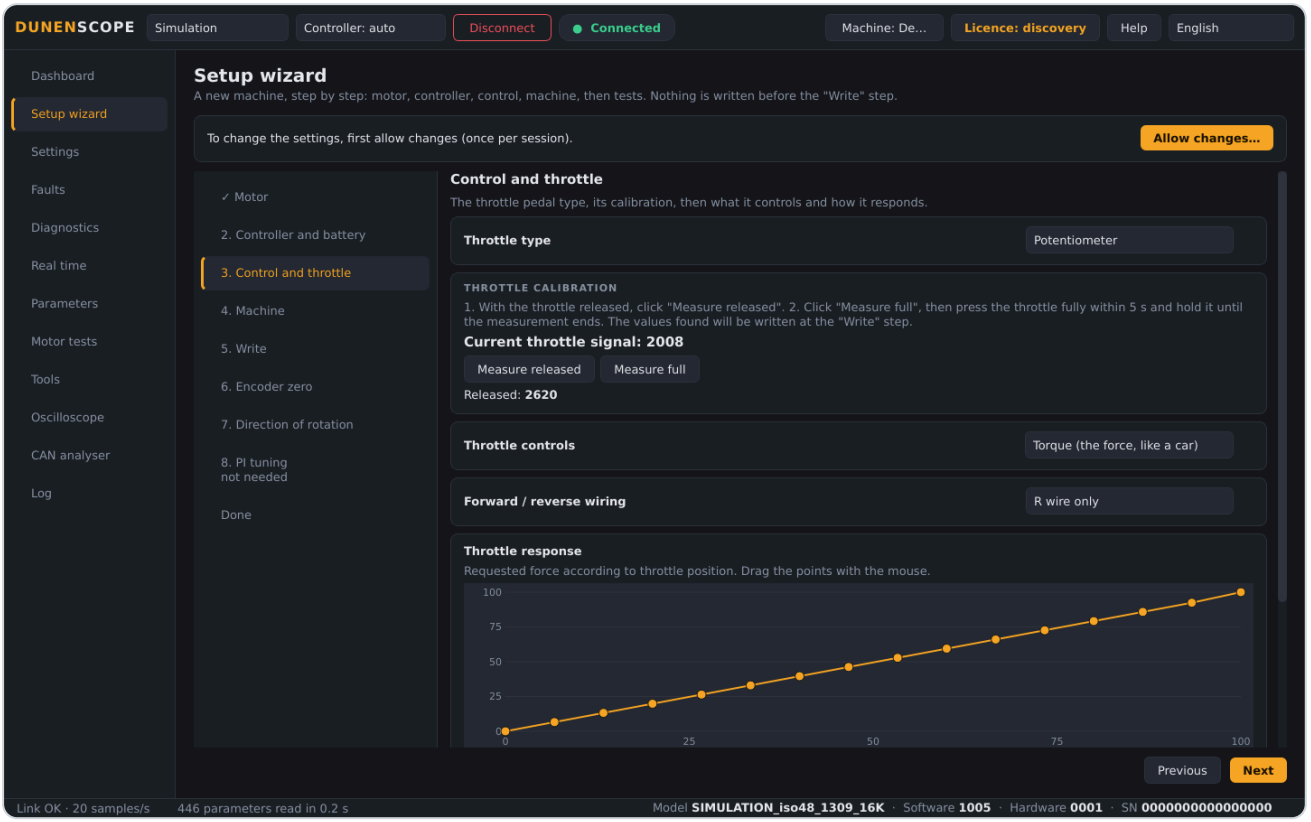
- **Four gauges**: motor speed with its setpoint, motor torque with its limit, battery voltage with the power-reduction and undervoltage cut-off thresholds, battery current with its limit. The scales follow the motor's capabilities.
- **Throttle** in %, with the raw signal (ADC) and the configured released and full-travel thresholds.
- **Estimated mechanical power**, and electrical power drawn from the battery.
- **Application tile**, depending on the machine type: hydraulic pump (speed and theoretical flow, as on the screenshot), vehicle speed in km/h, cable speed and pull of a winch, conveyor speed, fan or compressor speed after the gearbox, or electrical power.

- **State** (Running, direction, time powered), current **fault / warning**, motor and controller **temperatures** with their power-reduction threshold.
- **Current controller settings**: control mode, maximum speed, maximum torque, maximum battery current, undervoltage cut-off; "All settings →" opens the Settings page.
- **Banner** (while nothing is connected): reminds what to do, with "Connect" and "Try without hardware (simulation)".

The status bar shows "Link OK · n samples/s", or "Unstable link" in orange when requests go unanswered; the details (requests, errors, time per request) appear on hover.

6. Setup wizard

The **Setup wizard** tab guides the configuration of a new machine, step by step. "Next" and "Previous" move between steps; a finished step is ticked ✓. Nothing is written before the "Write" step.



Setup wizard

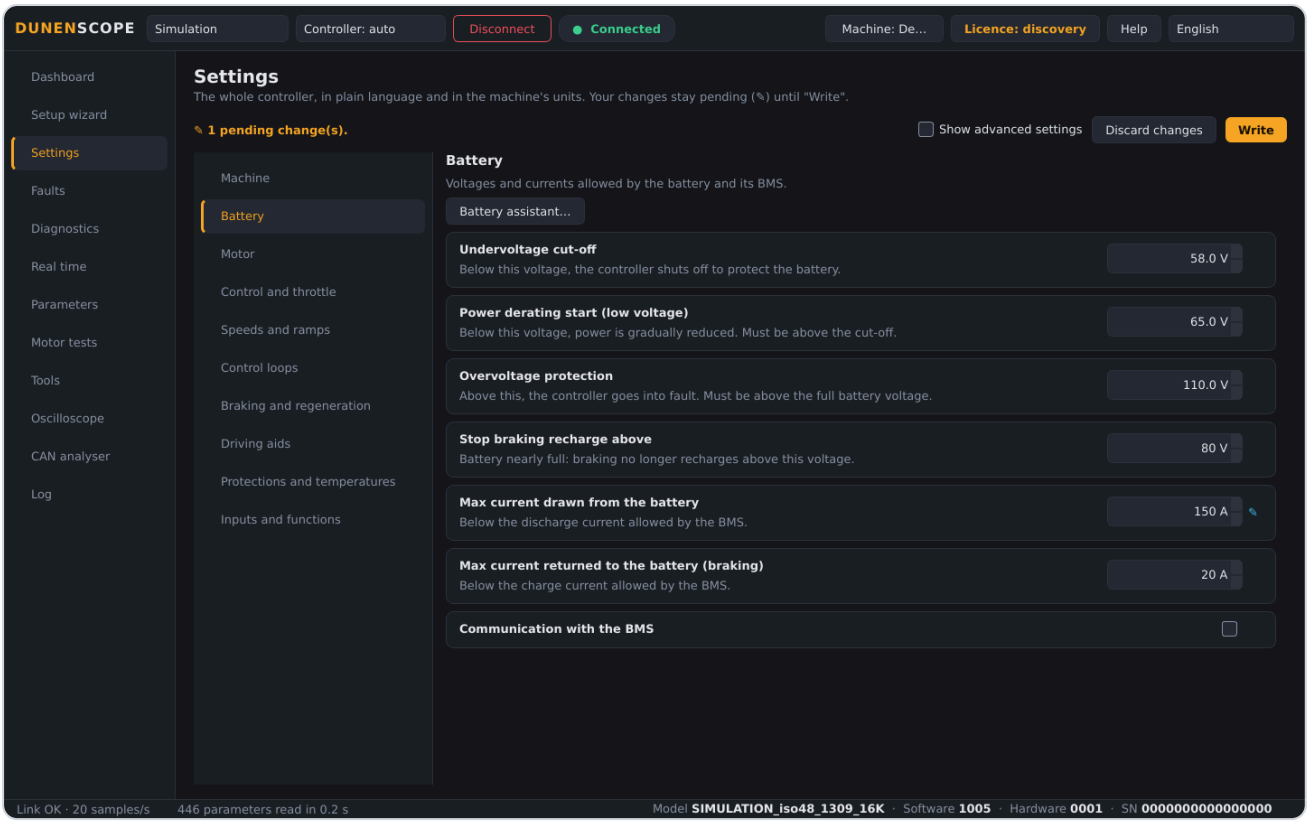
1. **Motor**: pole pairs, maximum torque, maximum phase current, temperature sensor and derating threshold, position sensor (type, pulses per revolution, direction).
2. **Controller and battery**: detected controller (model, versions), voltage thresholds, currents allowed by the battery and its BMS; "Battery assistant..." calculates them from the battery type and number of cells.
3. **Control and throttle**: pedal type, on-the-spot calibration ("Measure released", then "Measure full": a 5 s countdown leaves time to press the pedal fully, then the measurement runs; the signal is shown live), what the pedal controls (torque or speed), forward / reverse wiring, pedal response and torque ramp smoothness.

4. **Machine**: machine type, reduction ratio, wheel, drum or pump (kept on this PC), then maximum speed in the machine's units (with its equivalent in motor rpm) and, when the pedal controls speed, the times to reach it and to come back to a stop.
5. **Write**: list of pending changes in plain words, then "Write to the controller" (full backup before, read-back after).
6. **Encoder zero**: motor uncoupled from its load, the controller turns it slowly to find the zero of its position sensor; DunenScope says whether the zero was found, then "Keep this zero in the controller" saves it. If you move on without keeping it, DunenScope offers to.
7. **Direction of rotation**: a slow 5 s forward rotation; "Yes, right direction" or "No, reverse the direction" (the direction is then reversed and written: just repeat the test). The question is asked even if you stopped the test with STOP, as long as the motor turned; if it did not turn, DunenScope says so.
8. **PI tuning**: only when the pedal controls speed (step greyed out otherwise). With the motor mounted on the machine with its real load, "Auto-tune the PI..." opens the Motor tests tab; otherwise, skip the step and tune the PI later.
9. **Done**: summary (✓ done, ○ not done, or not needed) and final check: "Disconnect", key off for 10 s, key on, then "Connect". DunenScope reads the settings again and compares them with what was written: "✓ n setting(s) written, kept", or the list of those lost, prepared again, with a button to go back to the "Write" step. Connecting another controller resets the wizard.

Safety: the motor tests (steps 6 and 7) only start after the "Machine safe" box is ticked and once all changes are written; the STOP button and the Esc key cut the motor immediately, and each test stops by itself. In discovery mode, the wizard can be browsed and reads the values; changing them requires the full version ("View offers..." banner). Otherwise, the "Allow changes..." banner allows them, with confirmation.

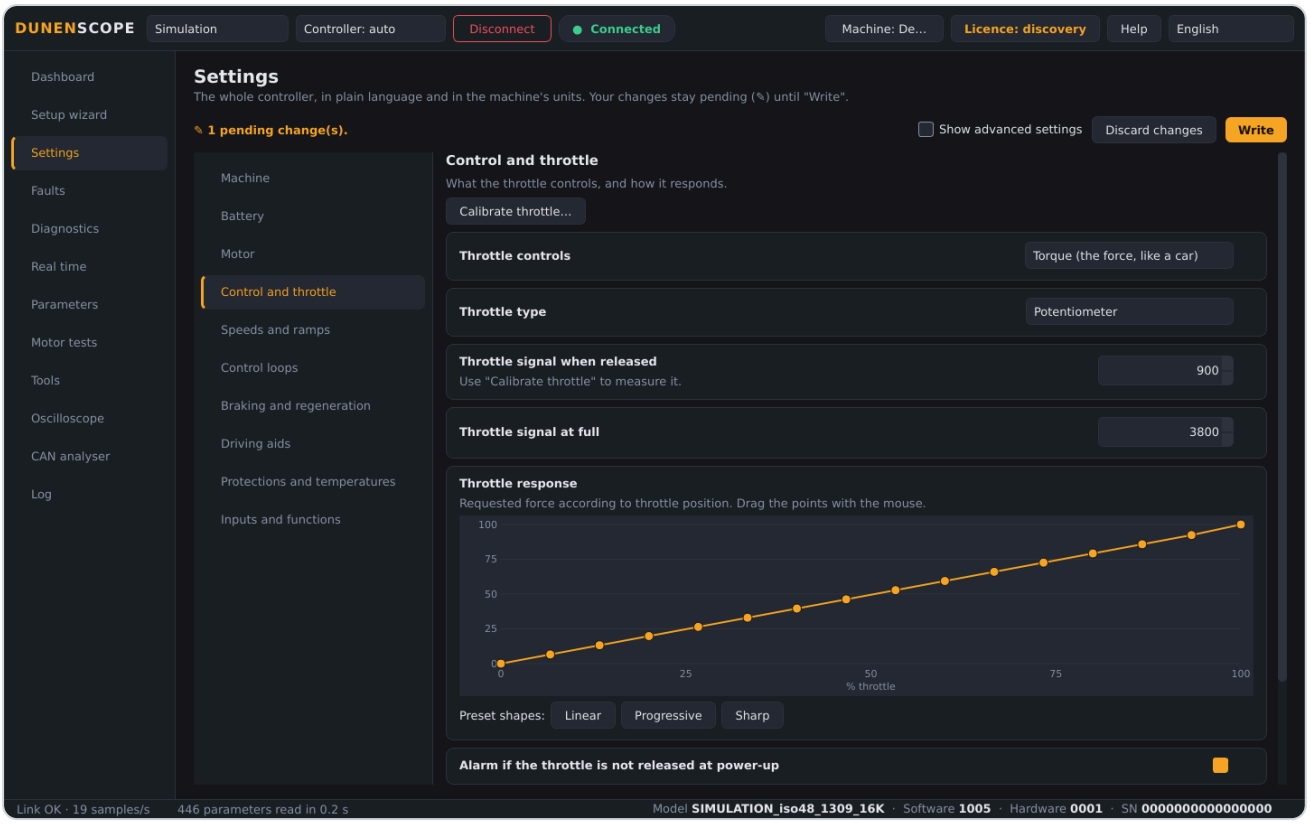
7. Settings

The **Settings** page shows the whole controller sorted by category, in plain language and in the machine's units. It is the simplest way to tune a machine: no need to know the parameter names.



Settings

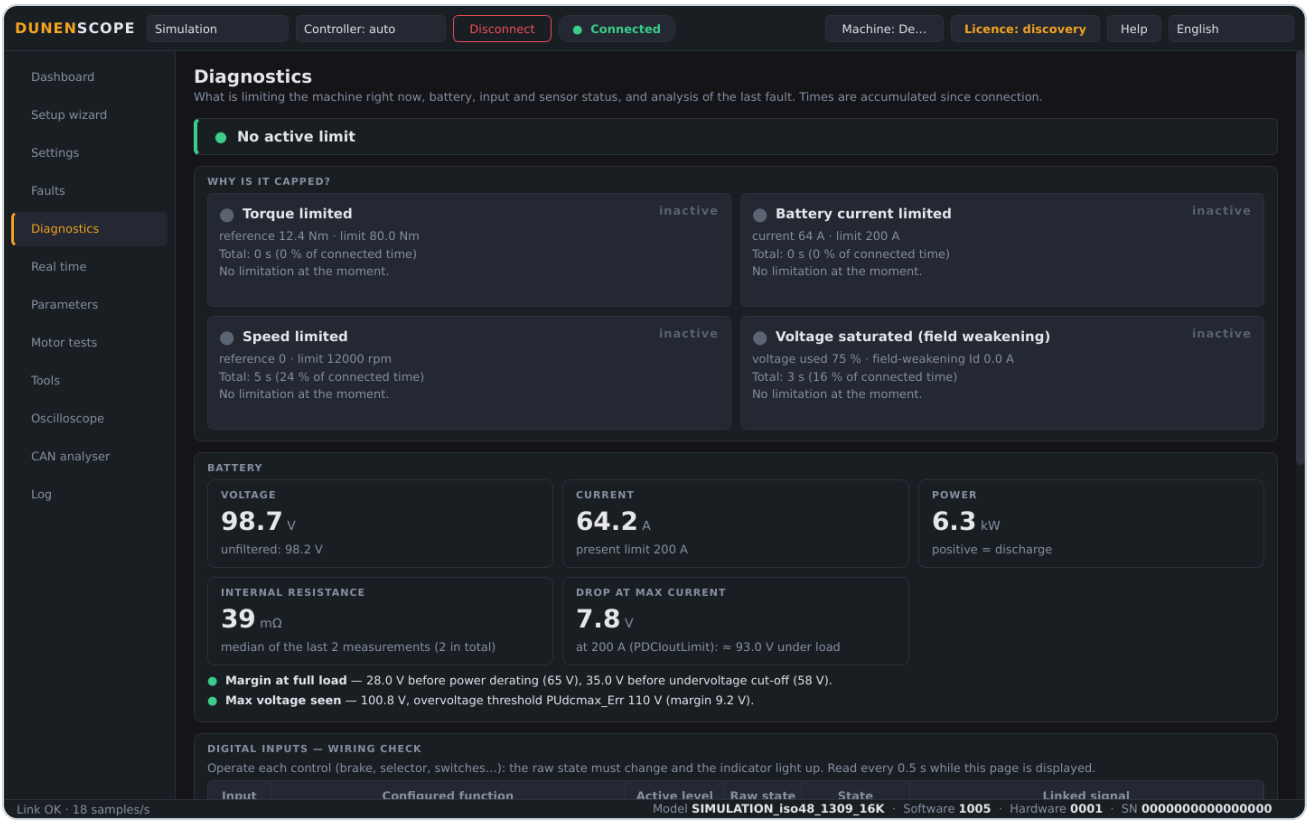
- **Categories**: Machine, Battery, Motor, Control and throttle, Speeds and ramps, Control loops, Braking and regeneration, Driving aids, Protections and temperatures, Inputs and functions.
- **Reading**: when the page opens, DunenScope reads the controller settings if this has not been done yet.
- **Machine units**: ramps in seconds (time to reach maximum torque or speed), coefficients in %, speeds in km/h (vehicle), L/min (pump), m/min (winch, conveyor) or rpm, depending on the machine type chosen in the Machine category.
- **Only what matters**: settings that are useless in the chosen mode are hidden (for example speed ramps when the throttle controls torque), as are those the connected model does not have. "Show advanced settings" shows the rest.
- **Curves**: throttle response and braking curves are changed by dragging the points with the mouse; "Linear", "Progressive" and "Sharp" give ready-made shapes.
- **Assistants**: "Battery assistant..." (easy setup), "Calibrate throttle..." (opens the "Control and throttle" step of the Setup wizard), "Encoder auto-learning...", "Auto-tune the PI...".
- **Nothing is written straight away**: every change is prepared in the parameter list (✎ rows). The first change asks you to allow changes (confirmation; full version for a real controller). "Write" goes through the usual steps: consistency check, full backup, writing, saving and read-back. "Discard changes" clears what was prepared.



Throttle response

8. Diagnostics

The Diagnostics page shows what is limiting the machine right now. A banner lists the active limits; on the screenshot, "No active limit".

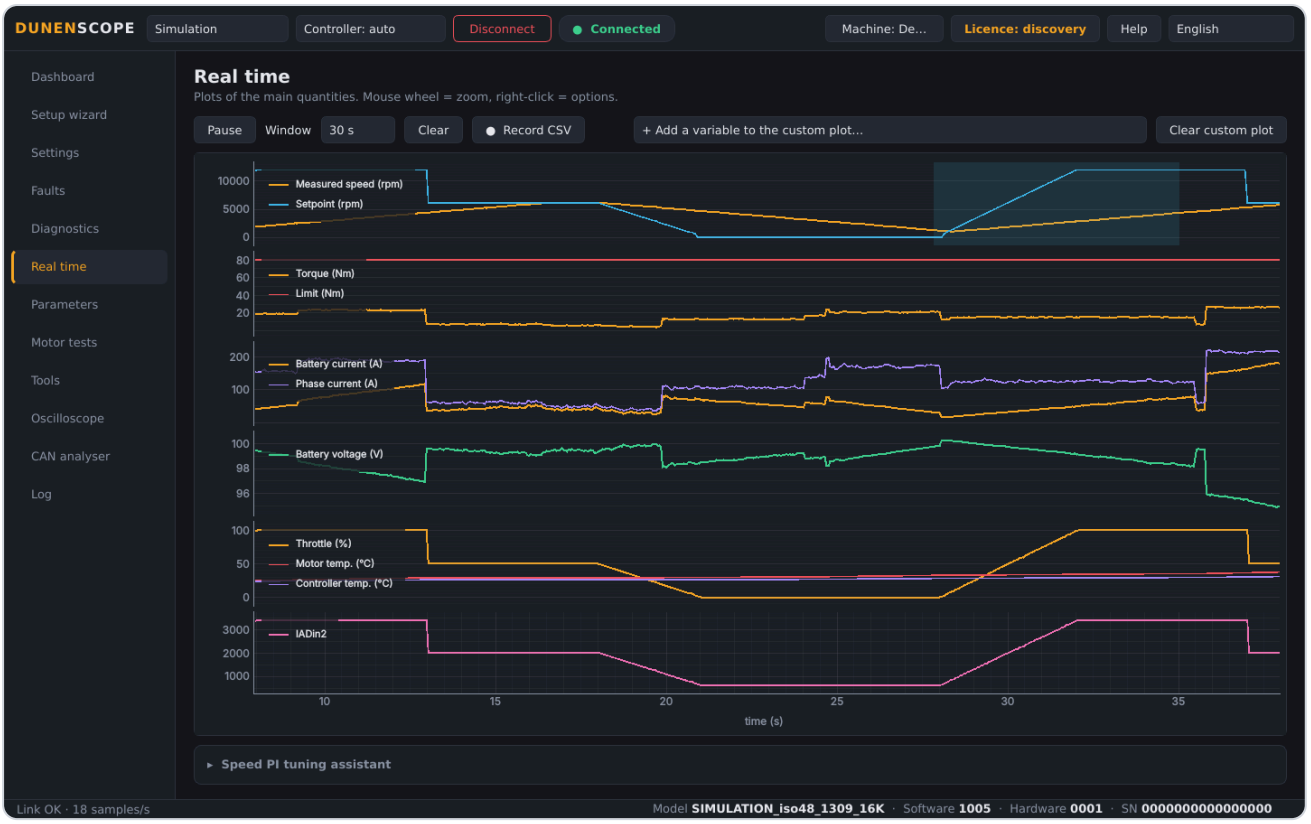


Diagnostics

- **Why is it capped?** Four cards in a 2×2 grid: "Torque limited", "Battery current limited", "Speed limited", "Voltage saturated (field weakening)". Each shows the reference and the limit, the time accumulated since connection and the cause.
- **Battery**: voltage, current, power, estimated internal resistance, expected voltage drop at maximum current, margin before power reduction and undervoltage cut-off, maximum voltage seen against the overvoltage threshold.
- **Digital inputs — wiring check**: configured function, active level and raw state of each input. Operate each control (brake, selector, switches…): the raw state must change.
- **Thermal**: motor and controller temperatures, raw motor sensor value, estimated time left before the power-reduction threshold.
- **Last fault analysis**: reads the conditions recorded at the last fault and gives the probable cause.

9. Real time

The Real time page plots the main quantities on six synchronised plots over an adjustable window (30 s on the screenshot). Mouse wheel = zoom, right-click = options.



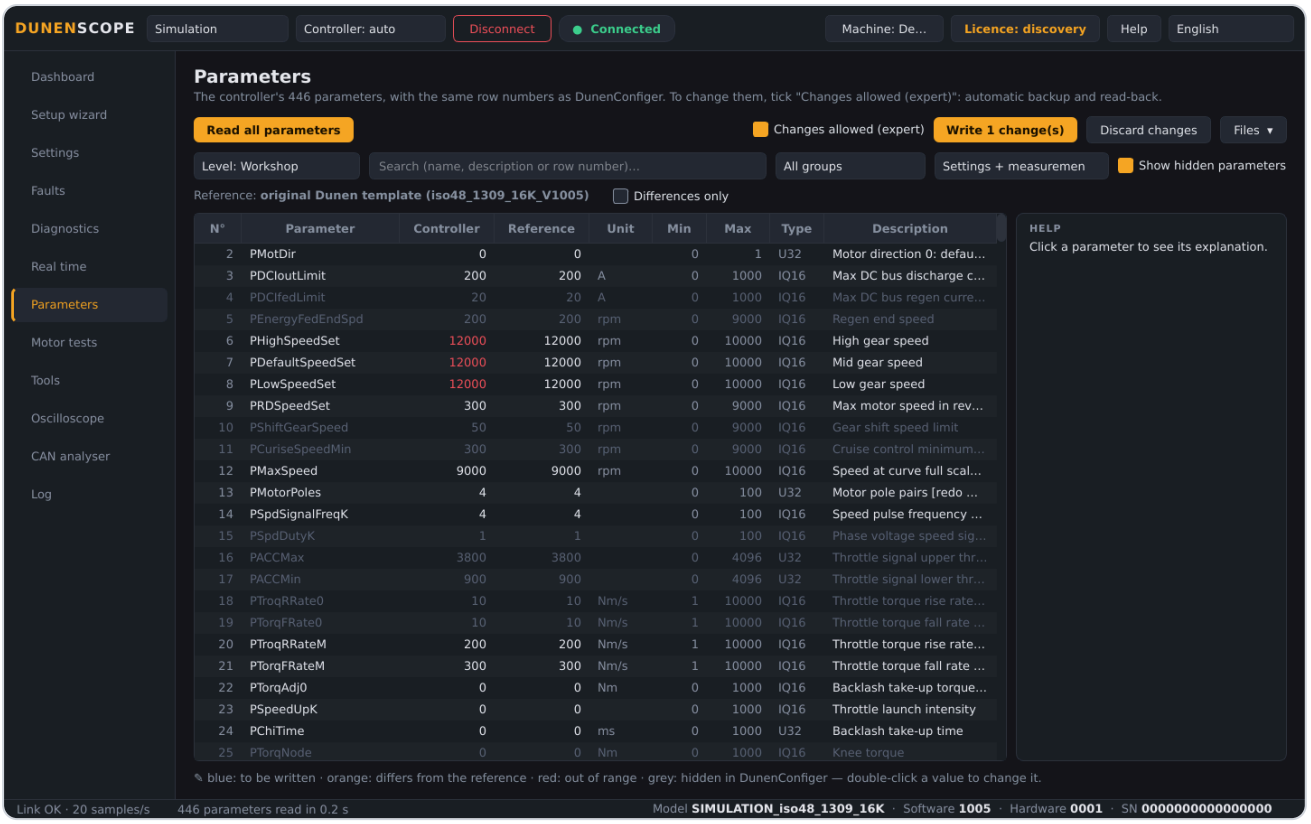
Real time

- **Plots**: measured speed and setpoint; torque and limit; battery current and phase current; battery voltage; throttle pedal and temperatures; custom plot.
- **Custom plot**: "+ Add a variable to the custom plot...", up to 4 variables of your choice (IADin2 on the screenshot). Double-clicking a row on the Parameters page also adds it.
- **Record CSV**: records the measurements to a file during the test. "Pause" freezes the display, "Clear" empties it.
- **Speed PI tuning assistant** (folded by default: click its title to open it): click "Measure the next setpoint change", press the pedal decisively (for example from half to full) and hold it for 3 seconds. DunenScope compares the measured speed with the setpoint and shows the lag on the ramp, overshoot, settling time, final error, ripple and number of oscillations, followed by a recommendation (for example "steady-state error: raise Ki by 20–50 %"). This assistant writes nothing to the controller.

To capture a short event (a fault, a jolt), use the Oscilloscope (chapter 18).

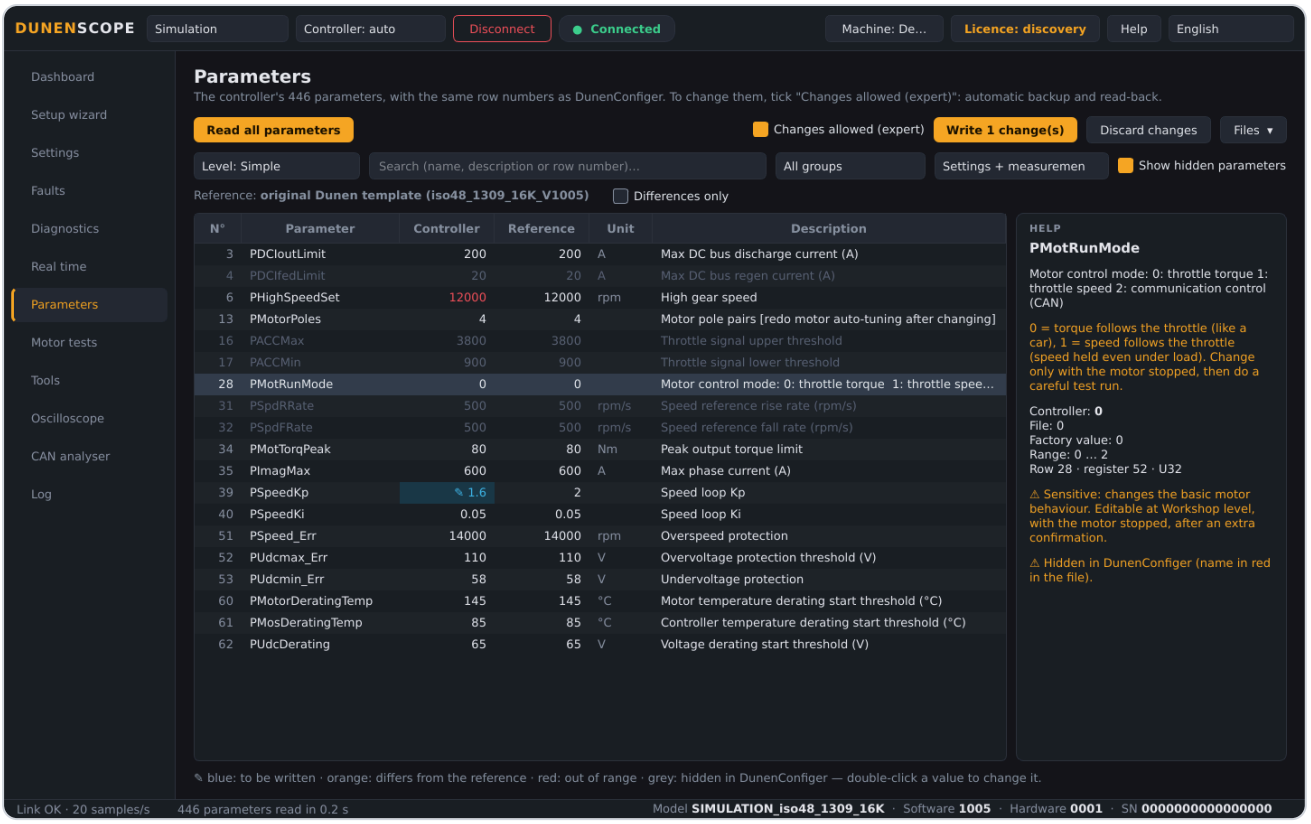
10. Parameters

The Parameters page lists the controller's 446 parameters, with the same row numbers as DunenConfiger. The screenshot shows changes allowed, with one pending change.



Parameters

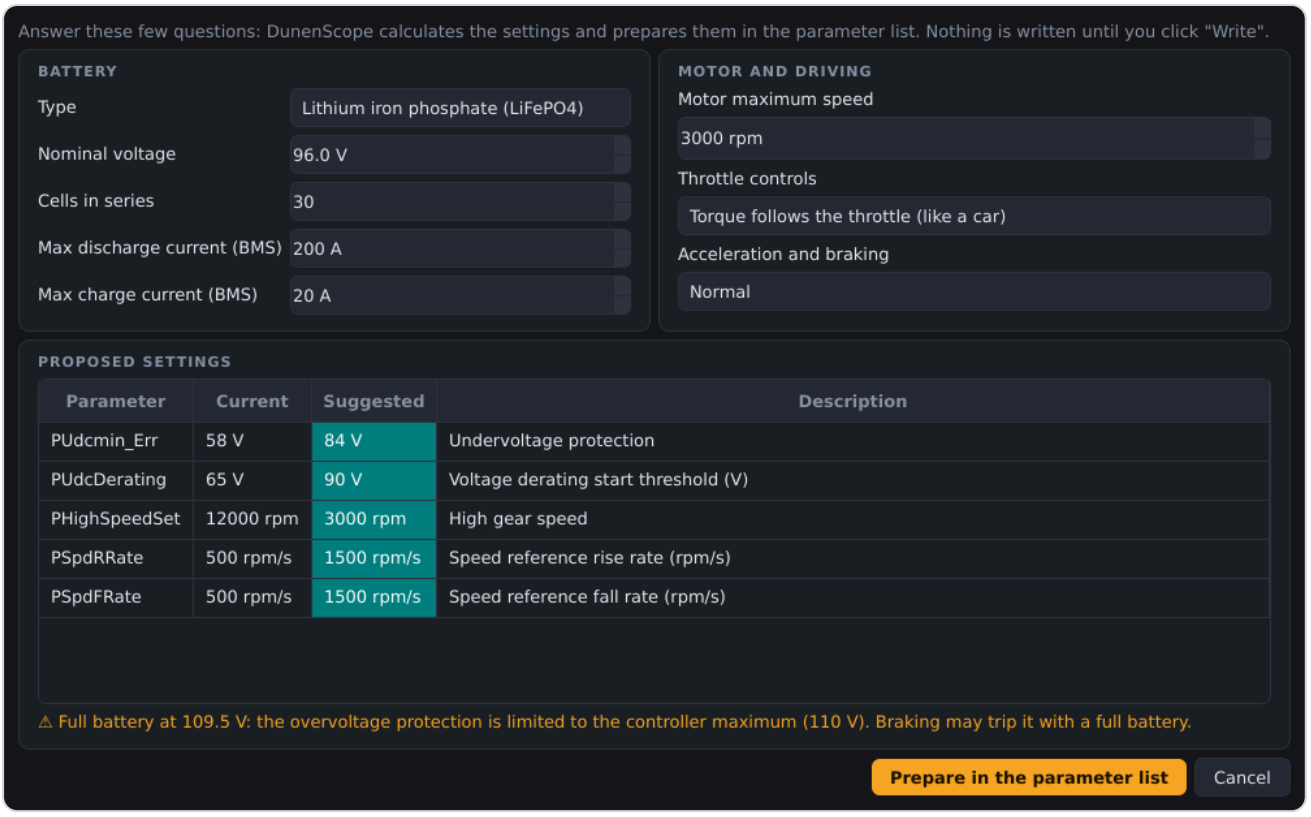
- **Read all parameters**: reads the full set; "Stop" interrupts the read.
- **Search and filters**: by name, description or row number; by group; settings and/or measurements; "Show hidden parameters" (greyed out); "Differences only".
- **Controller and Reference columns** side by side (the reference is the comparison file). The legend under the table explains the colours: blue ✎ to be written, orange different from the reference, red outside the file's Min / Max bounds (on the screenshot, 12000 rpm for a maximum of 10000), grey hidden in DunenConfiger. At the Simple level, the Min, Max and Type columns are hidden (the range is in the help panel).
- **Files menu**: choose the parameter file, compare with a file, export to xlsx (re-importable into DunenConfiger) or CSV.
- **Double-click the controller value**: change it directly. If needed, DunenScope offers to allow changes (with confirmation) and, for a sensitive setting shown at the Simple level, to switch to the Workshop level. Double-click elsewhere on the row: adds the variable to the custom plot on the Real time page.
- **Level**: *Simple* (default) shows only the twenty or so settings that are really changed; settings that affect the basic behaviour of the motor are only displayed there. *Workshop* shows every parameter. *Manufacturer* (special manufacturer key) also unlocks the parameters reserved for the manufacturer.
- **Help**: clicking a parameter shows, in the right-hand panel, its explanation, our advice, its current value, the file value, the factory value, the allowed range and the precautions.



Help

Easy setup

The "Battery assistant..." button (Controller and battery step of the Setup wizard, and Settings → Battery) asks a few questions: battery type and voltage (and number of cells), maximum discharge and charge currents allowed by the BMS, maximum speed, torque or speed control, soft, normal or sharp acceleration. DunenScope calculates the matching settings (undervoltage cut-off, power derating, overvoltage protection, current limits, overspeed, ramps, mode) and shows "current → proposed", with warnings when a value exceeds what the controller supports.



Easy setup

"Prepare in the parameter list" places the values in the list (rows marked ✎): nothing is written before "Write".

Changes allowed (expert)

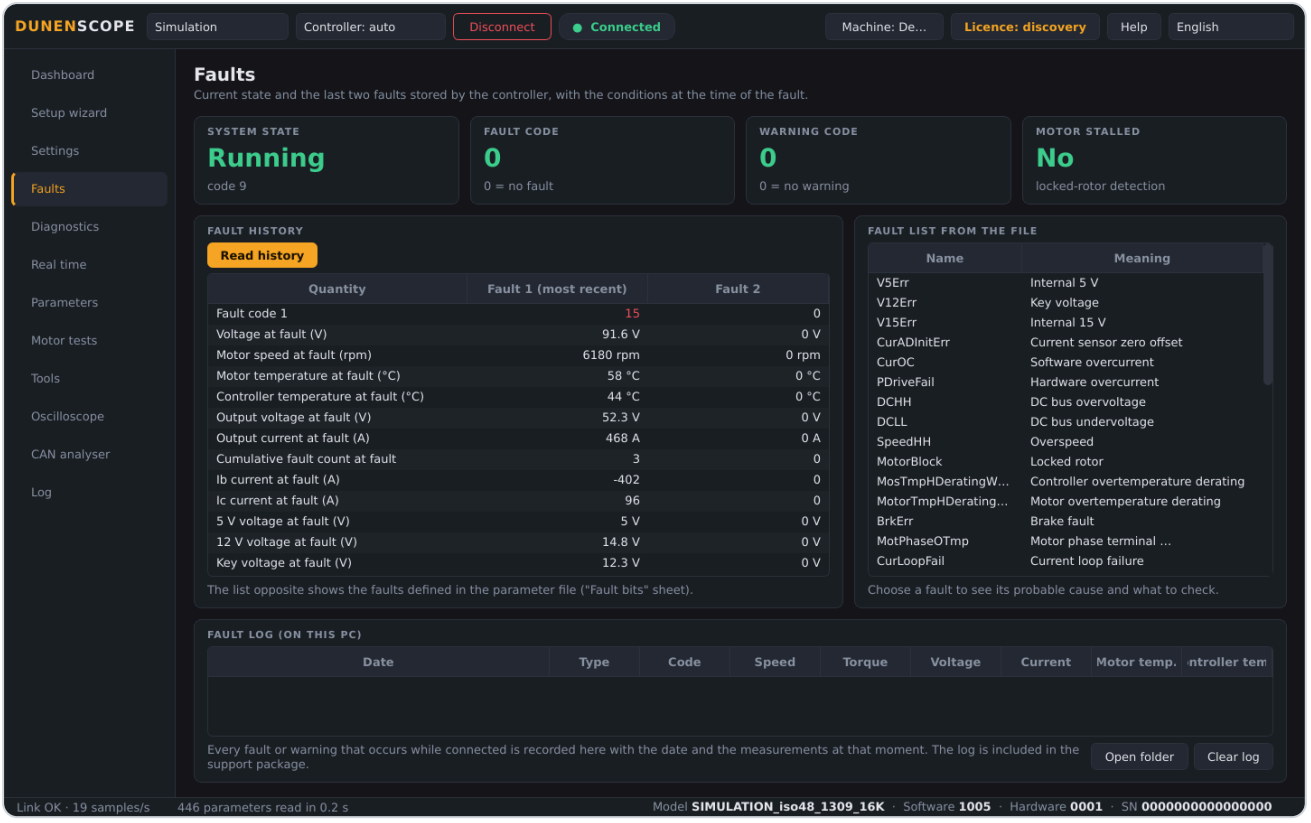
Tick "Changes allowed (expert)" and confirm. Values in the Controller column become editable. Each pending change is marked ✎ and counted on the "Write N change(s)" button; "Discard changes" drops them.

1. A value outside the file's bounds is rejected as soon as it is typed.
2. On "Write", DunenScope checks the consistency of the settings as they will be after writing: maximum speed above the overspeed threshold, throttle rest ≥ full, inconsistent voltage thresholds or thresholds too close to the current battery voltage, zero current or torque, zero speed gain in speed mode, derating temperatures too high... Serious problems (⛔) block the write at Simple level; at Workshop level, "Write anyway" must be confirmed. Points to check (⚠) are recalled in the confirmation.
3. DunenScope lists the changes and asks for confirmation, with a stronger warning for sensitive parameters (motor direction, pole pairs, encoder zero, protection thresholds...).
4. The motor must be stopped; otherwise the write is cancelled.
5. A full parameter backup is made just before. If it fails, nothing is written.
6. As with DunenConfiger, the whole settings table is rewritten, then the controller saves it in its permanent memory for 20 seconds: do not switch the key off before the final message. All settings are then read back for verification, and the log records each value as "written and verified".

Some parameters can never be changed: PBootNodeID and PBootCANBaudRat (CAN address and bit rate), PDefaultFn (restore defaults). Measurements (row 258 onwards) and factory calibration cannot be changed.

11. Faults

The Faults page shows the current state and the last two faults stored by the controller.



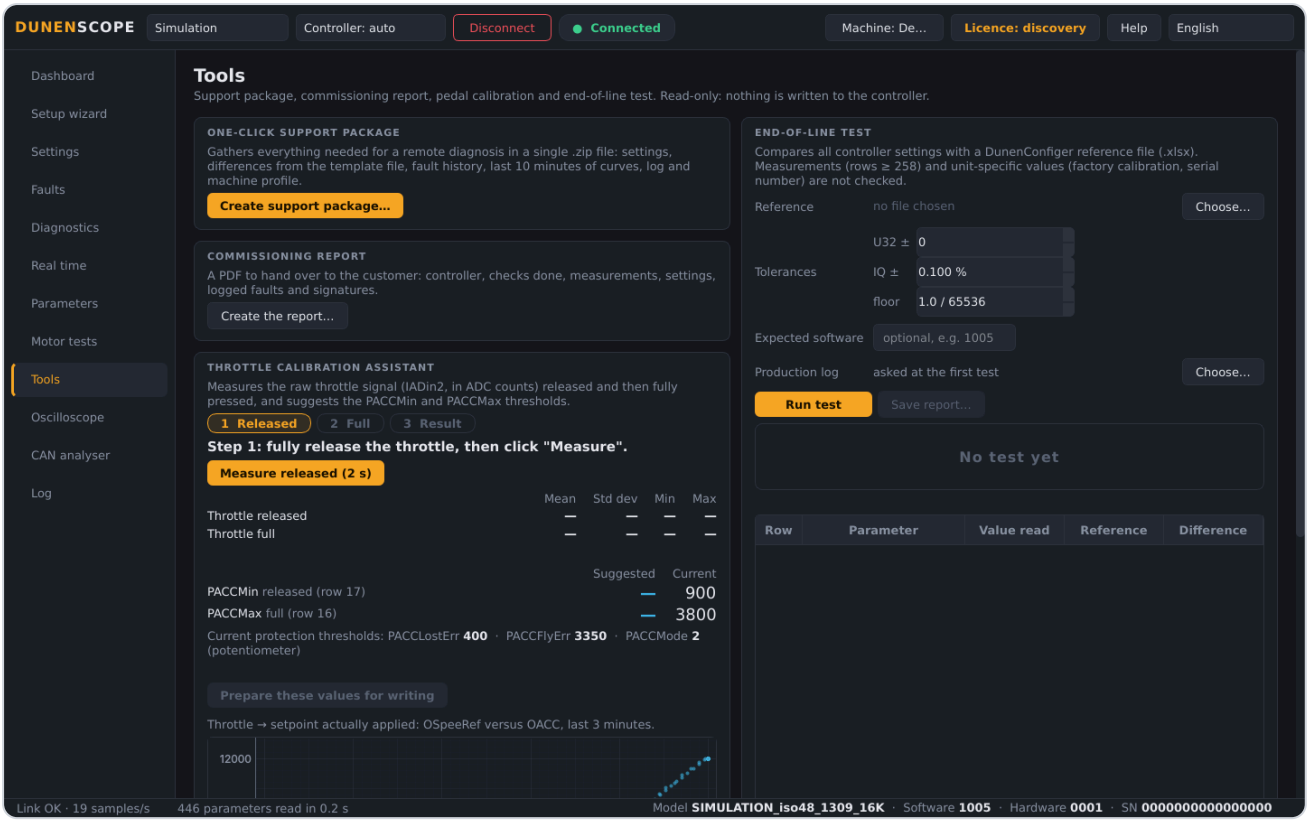
Faults

- **Top row**: system state, fault code, warning code, motor stall detection.
- **Fault history**: read automatically the first time the page is opened after connecting; "Read history" reads it again. It shows, for the most recent fault and the one before, the code and the conditions at the time of the fault: voltage, speed, temperatures, output voltage and current, Ib and Ic currents, 5 V, 12 V and key voltages, cumulative fault count.
- **Fault list from the file**: the faults defined in the parameter file ("Fault bits" sheet) and their meaning.
- **Fault log (on this PC)**: the controller only keeps its last two faults, without a date. DunenScope therefore records every fault or warning that appears while connected, with the date, the controller and the measurements at that moment (speed, torque, voltage, current, temperatures). The log (Documents\DunenScope\journal_defauts.csv) is included in the support package; "Clear log" resets it.

For a guided analysis of the last fault, use the Diagnostics page.

12. Tools

The Tools page groups the support package, the commissioning report, throttle calibration and the end-of-line test. It only reads from the controller.

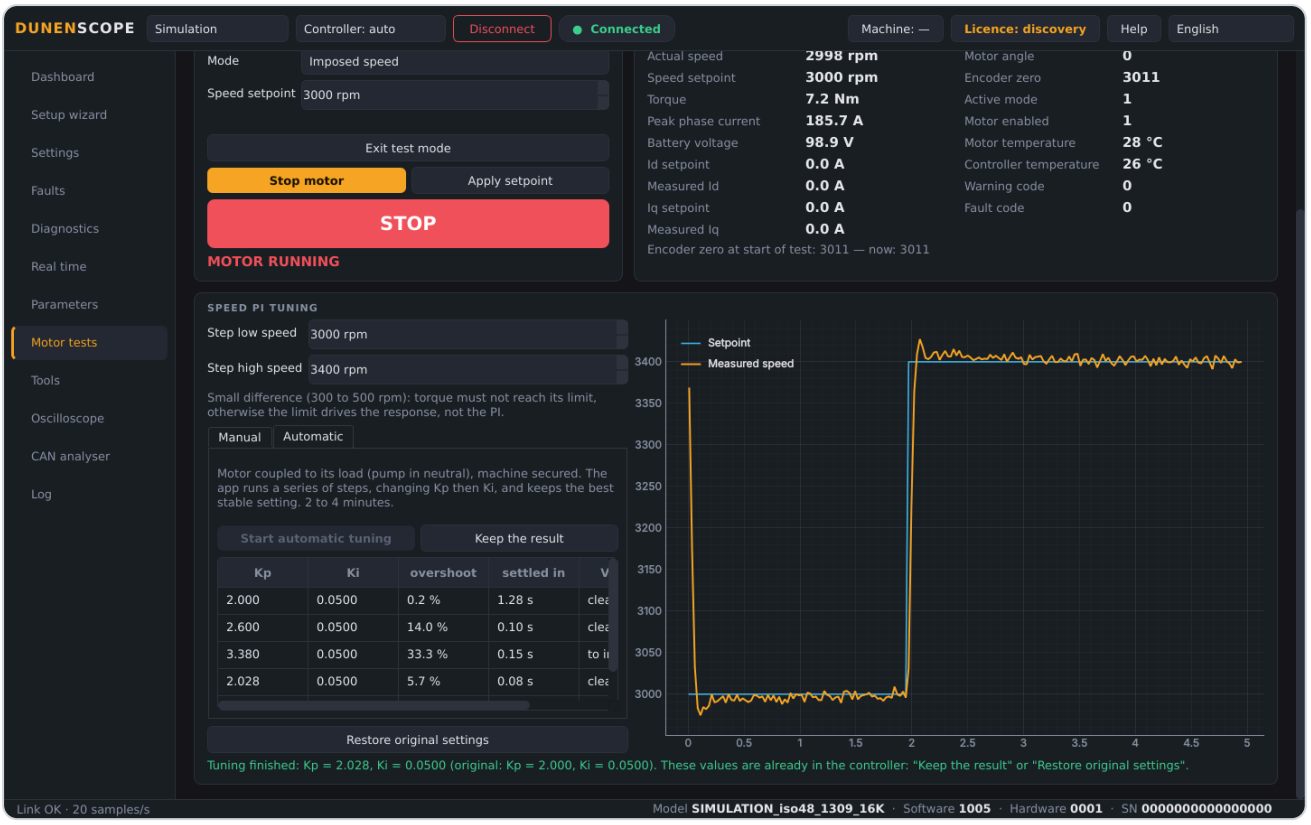


Tools

- **One-click support package**: "Create support package..." collects in a single .zip file the controller and machine information, all parameters, differences from the model file, fault history, plots of the last 10 minutes and the log. The customer sends it as is to support.
- **Commissioning report**: "Create the report..." produces a PDF to hand over to the customer (chapter 20).
- **Throttle calibration assistant**: in three steps (Released, Full, Result), it measures the raw pedal signal (IADin2) released then fully pressed and proposes PACCMin and PACCMax, next to the current values and the protection thresholds. "Prepare these values for writing" places them on the Parameters page, where they are written in expert mode. A plot shows the speed setpoint actually applied as a function of the pedal over the last 3 minutes.
- **End-of-line test**: compares every controller setting with a DunenConfiger reference file (.xlsx), with tolerances (U32, IQ in %, floor) and the expected software version. Measurements (row 258 onwards) and unit-specific values (factory calibration, serial number) are not checked. "Run test" shows the differences row by row; "Save report..." produces an HTML report (with a CSV version). Each test can be added to a production log.

13. Motor tests and PI tuning

The Motor tests page drives the motor directly from the PC, like the "AC motor debug" page of DunenConfiger. It requires expert mode, enabled on the Parameters page. The screenshot shows a completed automatic PI tuning.



Motor tests

Before starting: motor decoupled or machine made safe (wheels lifted, pump in neutral, nobody nearby), then tick "The machine is safe and the motor can turn freely".

- **Modes**: simulated throttle, imposed speed, imposed torque, encoder angle auto-learning, encoder adjustment (only when instructed by Dunen), current loop (Id / Iq).
- **Command**: "Enter test mode", "Start motor", "Apply setpoint". The **STOP** button and the **Esc** key stop the motor immediately.
- **Controller values** (debug variables), read live: actual speed and setpoint, torque, peak phase current, battery voltage, Id and Iq (setpoint and measured), motor angle, encoder zero, temperatures, warning and fault codes. The encoder zero at the start of the test stays displayed to check the auto-learning.

Speed PI tuning

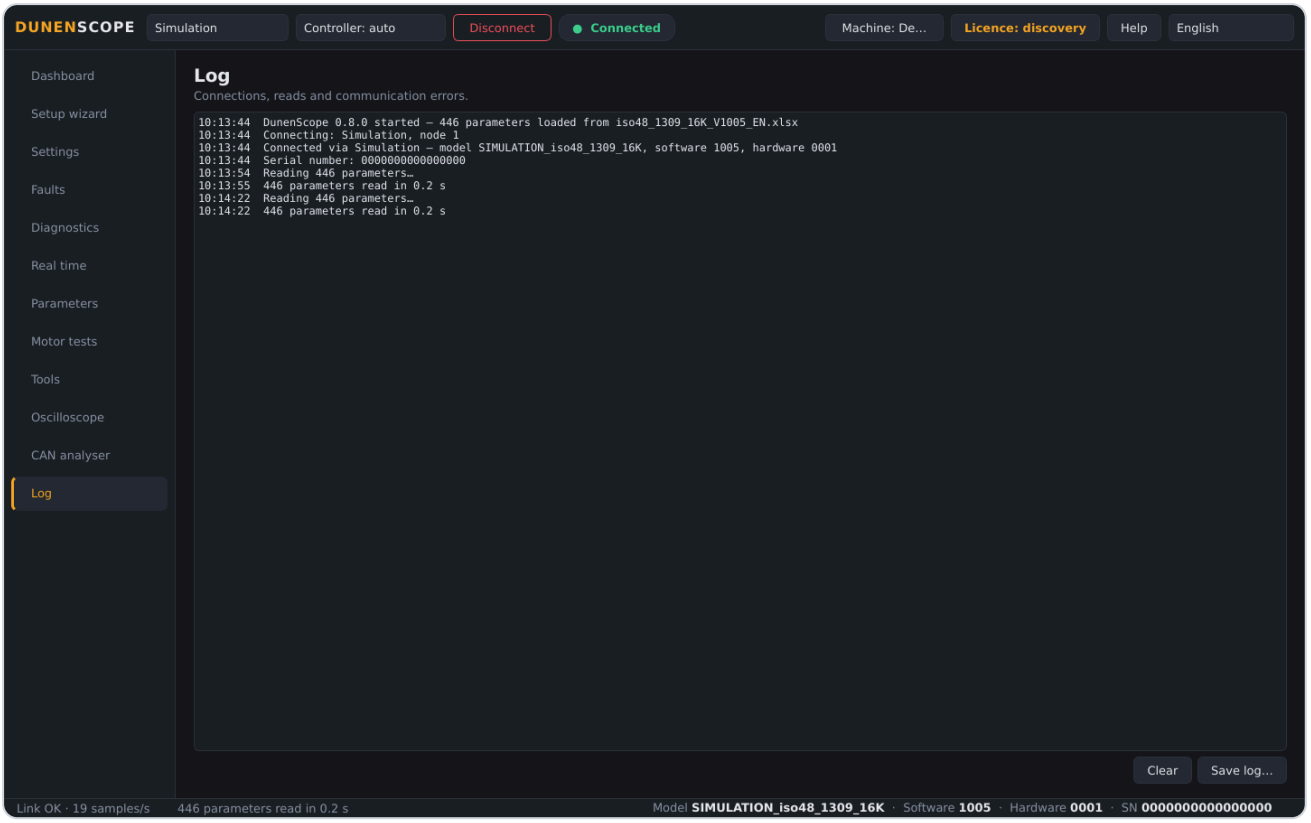
Tuning uses a speed step, for example 3000 → 3400 rpm. A small difference (300 to 500 rpm) keeps the torque below its limit: otherwise the limit drives the response, not the PI.

- **Manual**: enter Kp and Ki, write them, run a step and read the response and recommendations.

- **Automatic**: motor coupled to its load (pump in neutral), machine secured. DunenScope runs a series of steps, changing Kp then Ki, and keeps the best stable setting, in 2 to 4 minutes. Each trial is listed with Kp, Ki, overshoot, settling time and verdict.
- The plot overlays the setpoint and the measured speed. At the end, the status line gives the result and the original values (Kp, Ki). These values are already in the controller; choose "Keep the result" or "Restore original settings".

14. Log

The Log timestamps the session's events: start-up and loaded parameter file, connection (adapter, node, model, versions, serial number), reads, backups, verified writes, motor tests and communication errors.

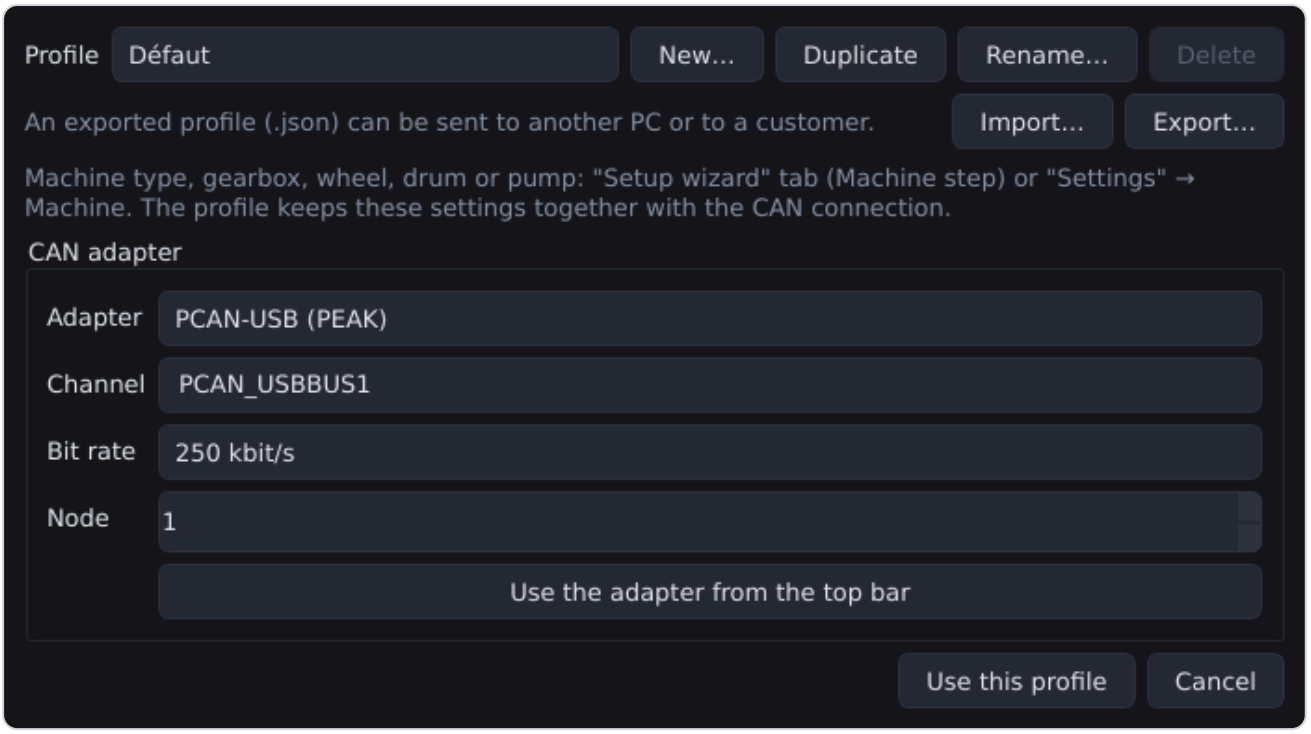


Log

"Save log..." saves it to a file to attach to a support request. "Clear" empties it.

15. Saved machines

The saved machines window opens from the "Machine: ..." button in the top bar. Each machine keeps its name, its description (set in Setup wizard → Machine or Settings → Machine) and the CAN adapter configuration.

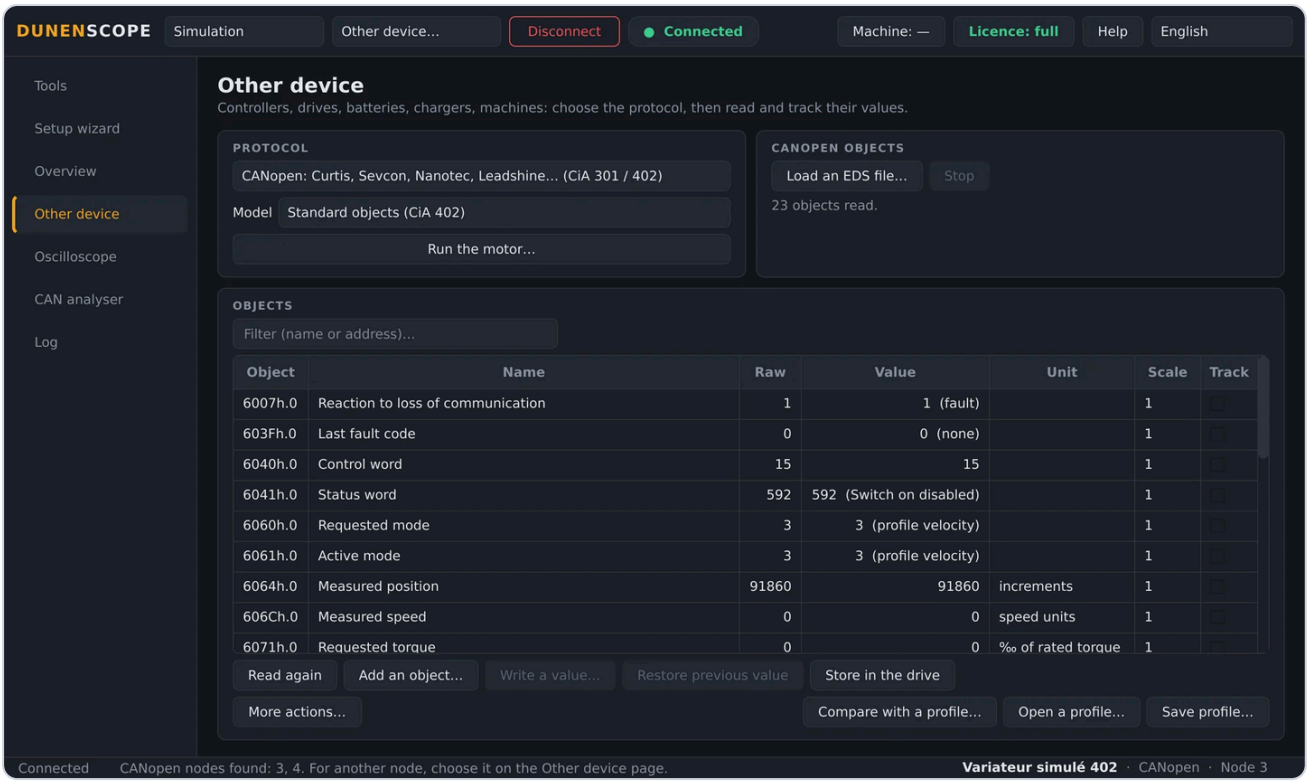


Saved machines

- **Machine**: described in the Setup wizard (Machine step) or in Settings → Machine: machine type (generic, vehicle, hydraulic pump, winch or hoist, conveyor, fan or compressor), reduction ratio (1 = direct drive), wheel or drum diameter, pump displacement. These values are not written to the controller: they show speeds in the machine's units.
- **CAN adapter**: adapter, channel, bit rate, node ("Automatic" by default). "Use the adapter from the top bar" copies the current configuration.
- **Management**: New, Duplicate, Rename, Delete. "Export..." produces a .json file to send to another PC or to a customer, who opens it with "Import...".

16. Other device

DunenScope is not only for Dunen controllers: it also reads controllers from other brands, industrial drives, batteries and chargers, machinery. Choose **"Other device..."** in the "Controller" list of the top bar. The left-hand menu then shows: Tools, Setup wizard (only for the devices it can set up), Overview, Other device, Oscilloscope, CAN analyser and Log.



Other device page: CANopen® drive, standard objects (CiA 402)

To try without hardware, choose the "Simulation" adapter: each protocol has its simulated device.

Protocols

The list of the **Protocol** card sorts the protocols into four families:

Family	Protocols	What you can do
Motor controllers	Dunen and compatible (Modbus over CAN)	search, read and write the registers of a Dunen not yet known
	Controllers compatible with VESC® Software, over USB or CAN (software 5.03 to 7.01)	measurements; motor and app settings, read and write; Setup wizard; run the motor; detect the motor
	Kelly (KLS-D, KAC-8080I, KBL / KHB over CAN; KLS / KAC over serial link), Votol EM (with or without its display), Fardriver, Bafang (M motors), ODrive	read only: measurements and faults
Industrial drives	CANopen® (CiA 301 / 402): Curtis, Sevcon, Nanotec, Leadshine...	read and write the objects; store; reset a fault; restart; factory settings; Setup wizard; run the motor. Read-only models: Curtis 1232E to 1239E, Sevcon Gen4, Nanotec PD4-E

Family	Protocols	What you can do
	Modbus RTU (RS-485, serial port), Modbus TCP (Ethernet)	search, read and write registers, coils and discrete inputs. Read-only models: Schneider Altivar ATV320, ASI BAC / Grin Phaserunner, Sabvoton / MQCON
Batteries and chargers	BMS over CAN, inverter protocol (Pylontech, Seplos, JK...); JK BMS; Daly BMS (CAN or serial link); JBD / Xiaoxiang / Overkill BMS; Pylontech battery over RS-485; Elcon / TC charger; Victron VE.Direct: BMV / SmartShunt battery monitor, solar charge controller or charger	read only: voltage, current, state of charge, cells, temperatures, alarms... depending on the device
Machinery and CAN bus	J1939 machine (engine, faults); DBC file (CAN bus listening)	listen only

Read only: no setting is changed; DunenScope sends at most read requests. *Listen only*: nothing is sent. The help line under the link settings recalls it, with how to connect the device.

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Connecting

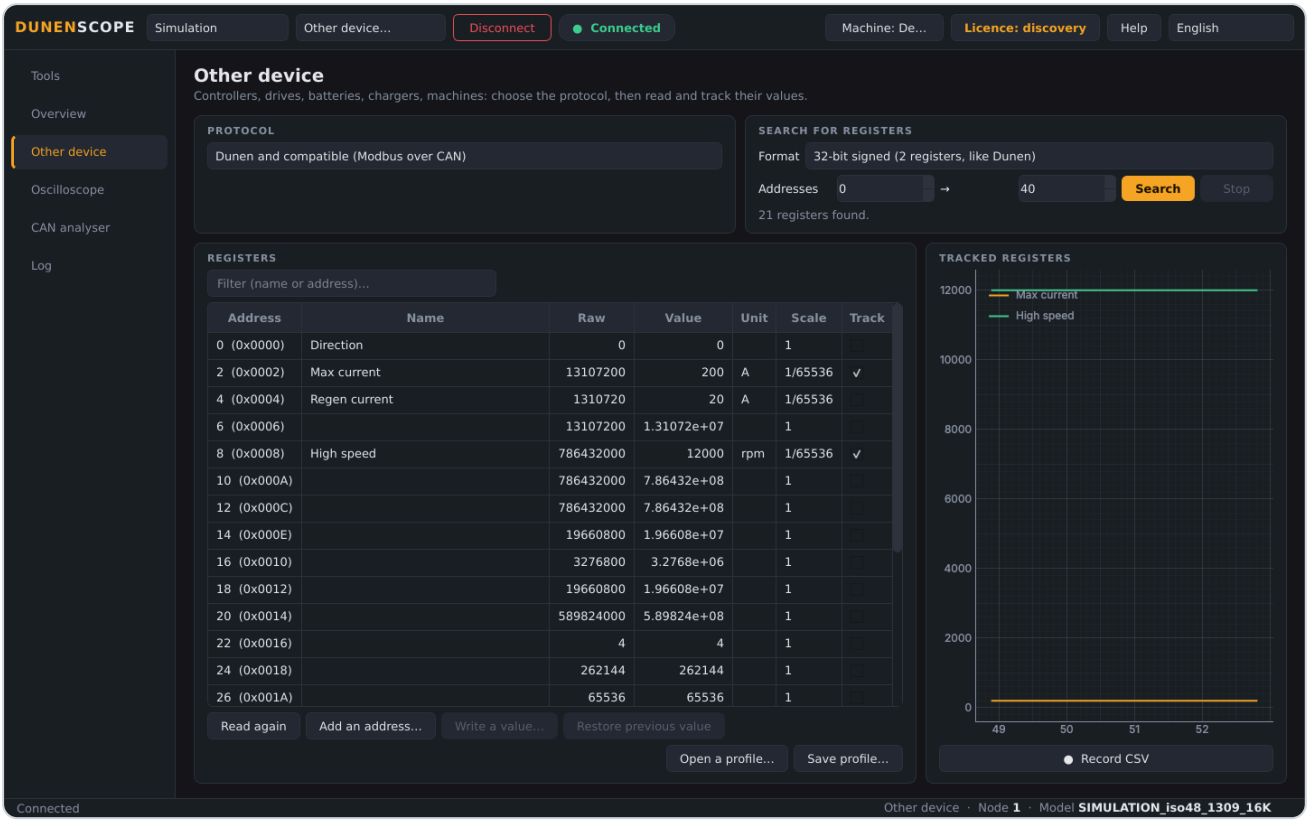
1. In the **Protocol** card, choose the protocol, then the **Model** if there is one (for a controller compatible with VESC® Software, the list is called **Show**: Measurements, Motor settings, App settings).
2. Set the link: CAN adapter from the top bar (bit rate and node "Automatic": DunenScope finds them by listening to the bus), serial port and its settings (Modbus RTU: often 19,200 baud, even parity, address 1), or IP address (Modbus TCP: usually port 502).
3. Click **"Connect"** in the top bar. The status bar shows the model, version and node.

Devices that broadcast (BMS, chargers, J1939, DBC file...) are listened to without transmitting anything, not even the frame acknowledgement. On a bench, a device alone with the adapter waits for this acknowledgement: tick **"Bench: device alone with the adapter"** (unticked at each start). IXXAT, CANable and USB-CAN Analyzer cannot listen without transmitting: choose the bit rate by hand.

Read, name, track

- **CANopen®**: the table shows the standard drive objects (status word, speed, torque, current, voltage...). **"Load an EDS file..."** (supplied by the manufacturer) gives all the drive's objects, with their names and types.
- **Modbus**: choose the function (holding registers 03, input registers 04, coils 01, discrete inputs 02), the format and the address range, then **"Search"**: nothing is written during the search.
- **Controllers compatible with VESC® Software**: the **"Show"** list chooses between measurements, motor settings and app settings. Settings with a plain-language name come first.
- **DBC file**: **"Load a DBC file..."** (supplied by the machine manufacturer): each signal of the file becomes a row. Without a file, an example is loaded.

- **J1939 machine**: engine speed, torque, voltage, temperatures, lit warning lamps and active faults (SPN, FMI), explained.
- **Batteries and chargers**: current counted positive when charging; the first 16 cells one by one, then the lowest, the highest and the difference.
- For all rows: the **"Filter (name or address)..."** field finds a row; give a name, a unit and a scale (fixed-point values: type 1/65536 for the Q16 format of Dunen controllers, 1/10 for tenths); tick **"Track"** to see the row on the chart, in the Overview (Modbus) and in the Oscilloscope. **"Record CSV"**, under the chart, records the tracked values to a file that Excel can read.



Modbus over CAN registers searched, named and tracked

Write (full version)

- **"Write a value..."** (or double-click the row): confirmation, old value kept, read-back. **"Restore previous value"** undoes it.
- **CANopen®**: a written value is lost at power-off until you click **"Store in the drive"**. **"More actions..."**: reset the drive fault, restart it, restore its factory settings (save a profile first to be able to go back).
- **Controllers compatible with VESC® Software**: the controller refuses settings from another version of its software, and limits the values itself to what its hardware allows.
- **Read-only model** (Curtis, Sevcon, Nanotec PD4-E, ATV320...): nothing is written while it is selected. To set up a Nanotec PD4-E, choose "Standard objects (CiA 402)" or its EDS file.

Profiles

"Save profile..." keeps the protocol, the link, the names and the values read. **"Open a profile..."** loads it again later: **"File on this PC..."**, or **"Online library..."** for profiles shared by other users (chapter 21). Sending a profile with the support package to the publisher lets this model be added to DunenScope.

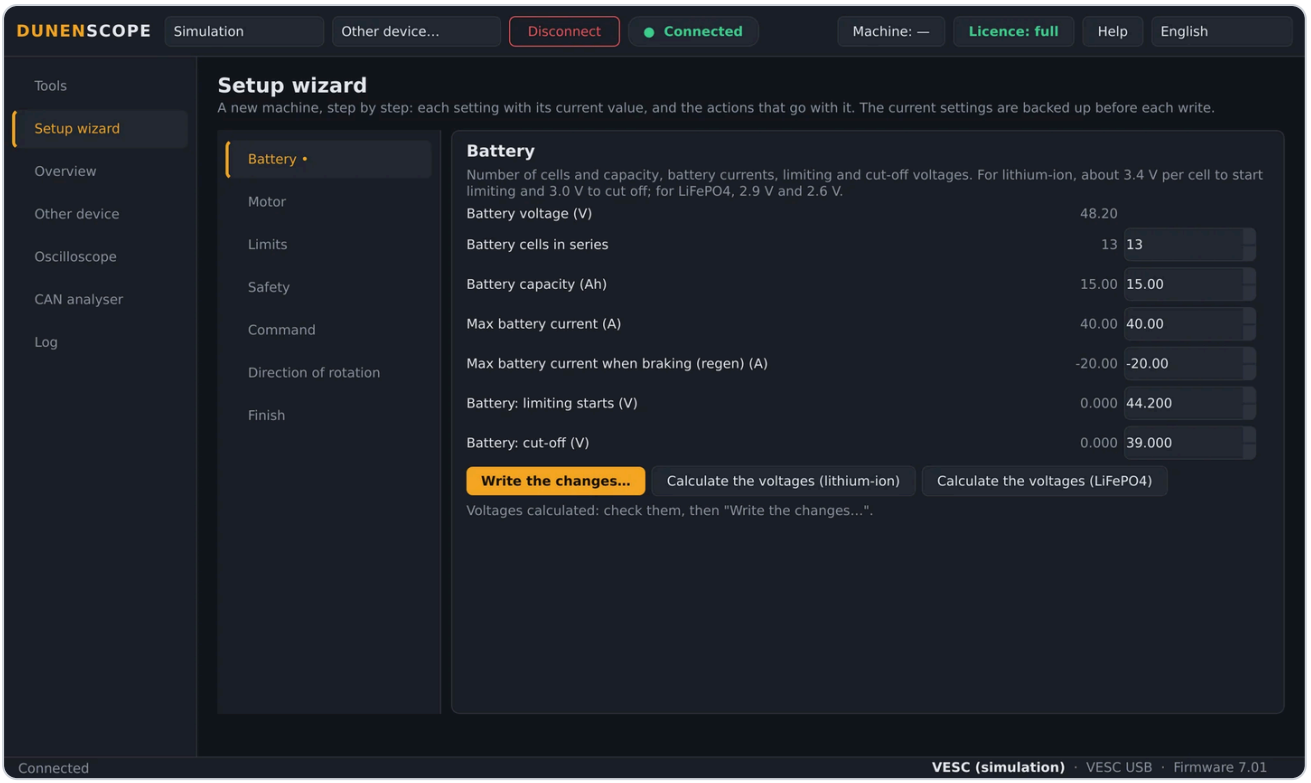
Automatic backups and comparison

Before the first write of each connection, and before any write of several values, a motor detection or a return to factory settings, DunenScope saves a backup profile (folder Documents\DunenScope\sauvegardes\autres_controleurs): the last values read in the table (all settings for a controller compatible with VESC® Software), the values read back of what is about to be overwritten, and the device they come from (model, number, software). The path is shown in the status bar and the log. If the backup fails, DunenScope asks whether to write anyway.

"Compare with a profile..." offers a profile, an automatic backup, or the latest backup of the connected device. DunenScope reads the settings back from the controller and shows those that differ from the profile; tick those to write, then **"Write the checked values to the controller..."**. Never written: command objects (enable, setpoints), the CAN number of a controller compatible with VESC® Software, objects unknown to the connected device, values not read back, out of format or of a different format. A profile from another device is flagged.

Guided setup

Setup wizard menu (controllers compatible with VESC® Software and CANopen® drives, except read-only models): a new machine, step by step. Each step shows its settings with their current value; change what is needed, then **"Write the changes..."** (confirmation, backup of the current settings, writing, read-back). A value entered but not yet written is kept from one step to the next (step marked •). Nothing is written during a motor test or a detection.



Setup wizard of a controller compatible with VESC® Software: Battery step

- **Controllers compatible with VESC® Software**: Battery (measured voltage; "Calculate the voltages (lithium-ion)" or "(LiFePO4)" proposes the limiting and cut-off voltages), Motor (poles, reduction, wheel, then "Detect motor..."), Limits (with the motor speed matching the maximum electrical speed), Safety (timeout without command, 1 to 1,000 ms), Command (input used and number on the CAN bus), Direction of rotation (low-speed test and reversal), Finish (profile and commissioning report).
- **CANopen® (CiA 402)**: Drive state (reset a fault, only if there is one), Motor (rating plate), Limits (with the equivalent in A or Nm of the ‰), Ramps, Safety (reaction to loss of communication), Direction and test (polarity), Save and finish (store in the drive, profile, report). Scales and units corrected on the Other device page are reused.

Run the motor

"Run the motor..." (Protocol card, once connected; controllers compatible with VESC® Software and CANopen® drives) opens the test window:

1. Make the machine safe (wheels lifted or transmission disengaged, nobody nearby) and tick "Machine safe: the motor can turn freely".
2. Choose the command: speed, current or duty cycle (compatible with VESC® Software); velocity or torque (CANopen®, drive units: check its manual).
3. **"Start motor"**, then change the value and **"Apply setpoint"** (or the Enter key).
4. **STOP** or the Esc key stop it at once. The motor also stops when the window is closed, on disconnection, after 2 minutes, and if DunenScope stops responding for one second.

Starting is refused if the CANopen® drive is in fault (reset it first), or if the "Timeout without command before stop" of the controller compatible with VESC® Software is not set between 1 and 1,000 ms. With CANopen®, a cut cable leaves the drive on its setpoint: keep the emergency stop within reach.

Detect the motor (compatible with VESC® Software)

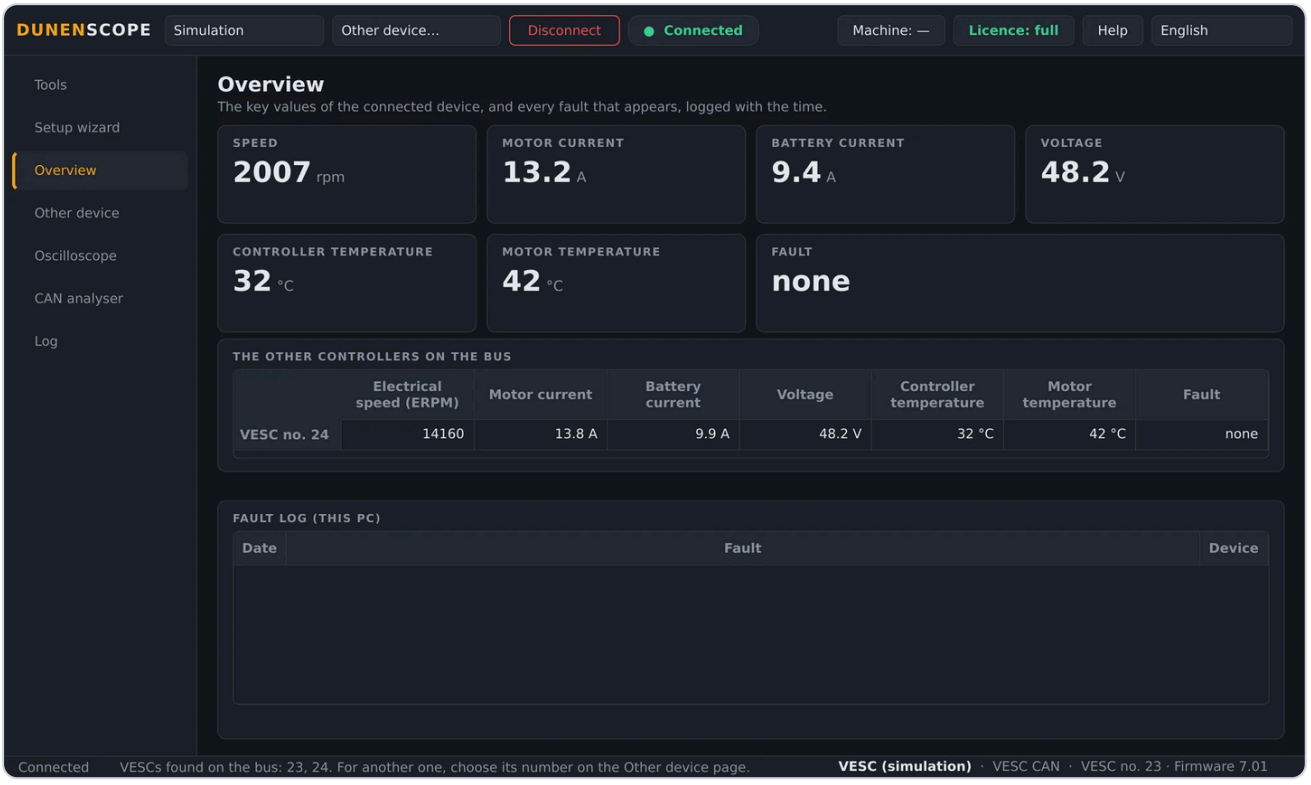
"Detect motor..." has the controller itself measure the motor:

- **"Measure R and L"**: the motor does not turn (a jolt is possible). DunenScope shows the measured resistance and inductance and proposes the current-loop gains; nothing is written before the window's write button.
- **"Start full detection"**: the motor turns by itself for up to one minute. The controller measures resistance, inductance and flux linkage, looks for the sensors (Hall or encoder), calculates the limits and gains, and saves everything. First choose the motor size and check the battery currents. During the measurement, DunenScope cannot interrupt it: keep your hand on the power cut-off.

The table, with its "Before" and "After" columns, shows what changed.

17. Overview and black box

Overview menu (Other device mode): the key values of the connected device, in large print, and every fault that appears.



Overview: connected controller compatible with VESC® Software and a second one on the same bus

- **Tiles**: speed, torque, currents, voltage, temperatures, state and fault; for a battery, the state of charge first, then the cells, the charge setpoint and the charge and discharge limits; for a solar charge controller, the panel power and voltage. With Modbus, these are the rows ticked "Track" on the Other device page.
- **The other controllers on the bus** (CANopen®, compatible with VESC® Software over CAN): at connection, DunenScope finds all the CANopen® drives and all the controllers compatible with VESC® Software on the bus. The same values are shown for each of the others, read every second.

To set up another controller, choose its node (or number) in the Protocol card and reconnect. To copy settings from one to another: "Save profile..." on the first, then "Compare with a profile..." on the second.

- **Fault log (this PC)**: each fault that appears is recorded with the time, the device (and its number for the other controllers on the bus) and the measurements at that moment. It is the same log as for Dunen controllers (Documents\DunenScope\journal_defauts.csv), included in the support package. Clicking a row shows the probable cause and what to check.
- **Black box**: for each fault, DunenScope keeps the measurements of the 20 seconds that preceded it, in a CSV file that Excel can read (negative time before the fault). The link "Black box: measurements of the seconds before the fault (CSV)" appears under the explanation of the fault. The files are in Documents\DunenScope\boite_noire; the 200 most recent are kept.

The values are read even when the page is hidden (more slowly): no fault escapes the log or the black box.

18. Oscilloscope

Oscilloscope menu (Dunen controllers and other devices): up to 4 quantities captured around an event, with what happened just before.



Oscilloscope: four channels captured over 2 s, cursors A and B

1. **Channel 1 to Channel 4**: choose the quantities. Dunen: the measurements of the parameter file (those of the dashboard first). Other devices: the Overview values and the rows ticked "Track".
2. **Trigger**: "immediate", "a fault appearing", or a quantity "rising above" or "falling below" a threshold.
3. **Duration** (1 to 60 s) and the part kept **Before the trigger** (10, 25 or 50 %).

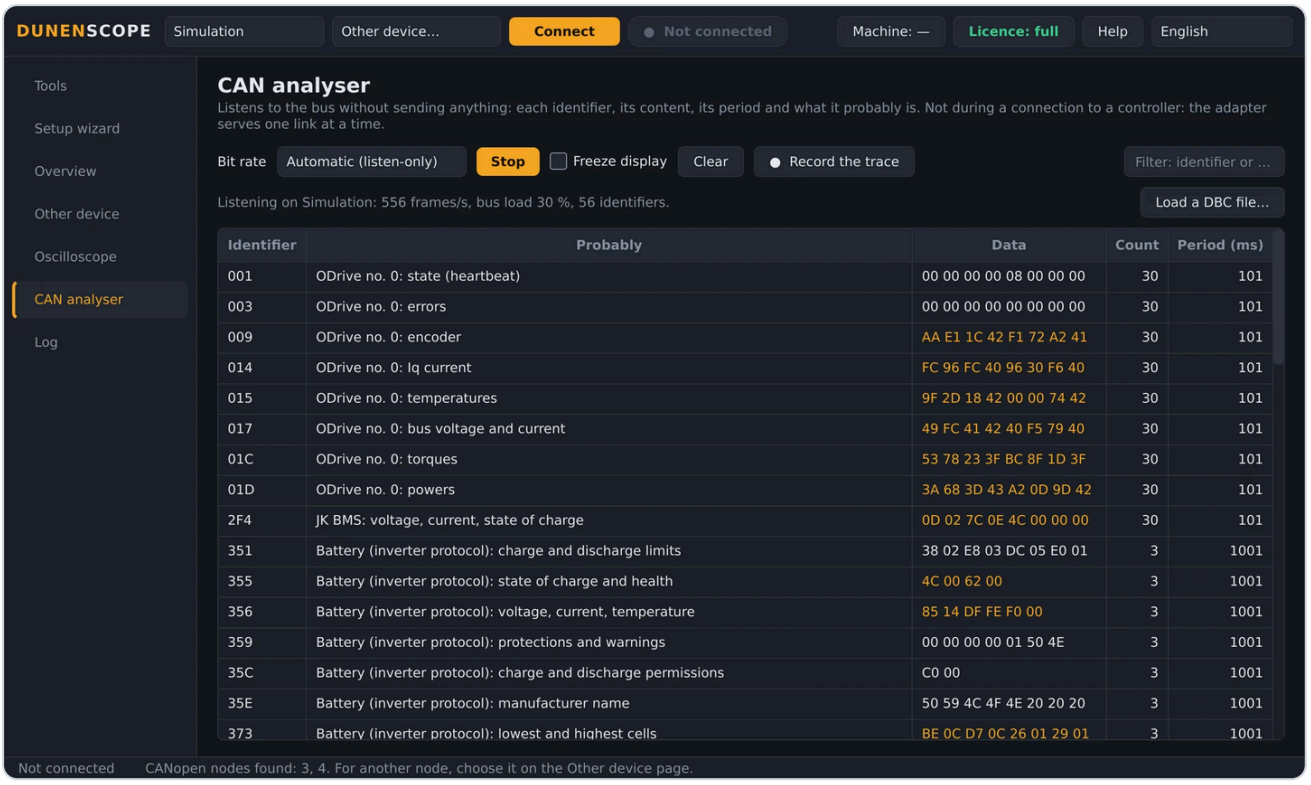
4. **"Arm"**. The status line says what is awaited and how many samples arrive per second; "Stop" abandons the capture.

The capture fills as fast as the controller answers: about 20 samples per second for a Dunen; for other devices, as fast as the link allows. It continues if you change page. The Motor tests and Setup wizard (Dunen) pages stop continuous reading: no samples meanwhile.

Once the capture is finished, drag cursors **A** and **B**: the time difference and the difference of each quantity are shown under the plots. **"Save CSV..."** keeps the capture (time counted from the trigger).

19. CAN analyser

CAN analyser menu: DunenScope listens to the bus without sending anything and shows each identifier, what it probably is, its content and its period.



CAN analyser: simulated bus (ODrive, JK BMS, inverter-protocol battery)

1. Disconnect from any controller: the adapter serves one link at a time. The analyser also stops by itself when you connect.
2. Choose the **Bit rate**, or "Automatic (listen-only)": DunenScope listens at each possible bit rate before opening the adapter, because a wrong bit rate would disturb the bus.
3. **"Listen"**. **"Stop"** ends listening.

While listening:

- **Columns**: Identifier, Probably, Data (bytes that change in colour), Count, Period (ms). The status line gives the frames per second, the bus load and the number of identifiers; error frames cast doubt on the bit rate.

- **Probably**: CANopen® (NMT, SYNC, EMCY emergencies, PDO, SDO, heartbeat), compatible with VESC® Software, Kelly, Dunen, J1939 (known messages named, with their PGN), batteries, chargers, ODrive...
- **"Load a DBC file..."**: name of each message of the file and value of its signals. Once loaded, the same button offers another file, "Only the messages of the file" and "Remove the DBC file".
- **"Freeze display"**, **"Clear"**, and the "Filter: identifier or text" field.
- **"Record the trace"** writes each frame to a CSV file (Excel).

The adapter is opened listen-only, without transmitting, not even the acknowledgement: PEAK, Kvaser, Vector, candleLight (if it announces it) and DunenScope Link. The others (IXXAT, CANable, USB-CAN Analyzer) cannot stay silent and are not opened. With the "Simulation" adapter, the analyser listens to a simulated bus.

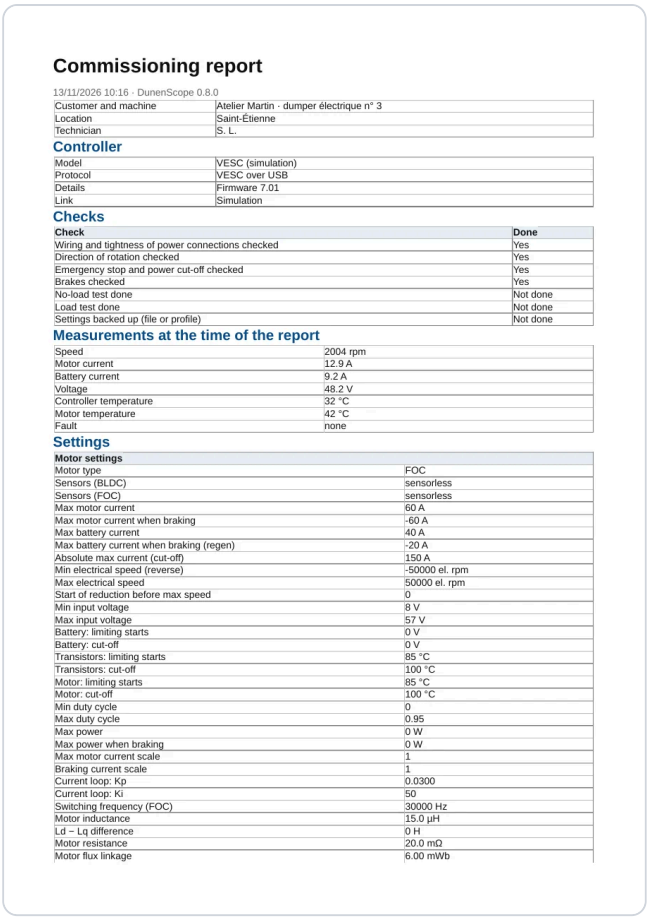
20. Commissioning report

Tools page → **"Create the report..."** (or last step of the Setup wizard of another device, "Create the commissioning report..."): a PDF to hand over to the customer, for all controllers.

1. Enter **Customer and machine**, **Location** and **Technician** (kept for next time).
2. Tick the checks done: wiring and tightness of power connections, direction of rotation, emergency stop and power cut-off, brakes, no-load test, load test, settings backed up (file or profile).
3. Add **Remarks** if needed, then **"Save the PDF..."**.

The PDF (A4) also includes the connected controller (model, protocol, versions, serial number, link), the measurements at the time of the report, the settings read (Dunen: those of the Settings page, read if needed; compatible with VESC® Software: motor and app settings; others: the rows of the table), the faults logged for this controller (with their probable cause for other devices) and two boxes for the technician's and the customer's signatures.

With no controller connected, DunenScope offers to create the report without settings or measurements. If the link changes while you type, the report uses the values read when the window was opened.



Commissioning report

13/11/2026 10:16 · DunenScope 0.8.0

Customer and machine	Atelier Martin · dumper électrique n° 3
Location	Saint-Étienne
Technician	S. L.

Controller

Model	VESC (simulation)
Protocol	VESC over USB
Details	Firmware 7.01
Link	Simulation

Checks

Check	Done
Wiring and tightness of power connections checked	Yes
Direction of rotation checked	Yes
Emergency stop and power cut-off checked	Yes
Brakes checked	Yes
No-load test done	Not done
Load test done	Not done
Settings backed up (file or profile)	Not done

Measurements at the time of the report

Speed	2004 rpm
Motor current	12.9 A
Battery current	9.2 A
Voltage	48.2 V
Controller temperature	32 °C
Motor temperature	42 °C
Fault	none

Settings

Motor settings	
Motor type	FOC
Sensors (BLDC)	sensorless
Sensors (FOC)	sensorless
Max motor current	60 A
Max motor current when braking	-60 A
Max battery current	40 A
Max battery current when braking (regen)	-20 A
Absolute max current (cut-off)	150 A
Min electrical speed (reverse)	-50000 el. rpm
Max electrical speed	50000 el. rpm
Start of reduction before max speed	0
Min input voltage	8 V
Max input voltage	57 V
Battery: limiting starts	0 V
Battery: cut-off	0 V
Transistors: limiting starts	85 °C
Transistors: cut-off	100 °C
Motor: limiting starts	85 °C
Motor: cut-off	100 °C
Min duty cycle	0
Max duty cycle	0.95
Max power	0 W
Max power when braking	0 W
Max motor current scale	1
Braking current scale	1
Current loop: Kp	0.0300
Current loop: Ki	50
Switching frequency (FOC)	30000 Hz
Motor inductance	15.0 µH
Ld – Lq difference	0 H
Motor resistance	20.0 mΩ
Motor flux linkage	6.00 mWb

First page of a commissioning report (PDF)

21. Online library

The online library gathers profiles shared by other users: the names, units and scales of a model's rows. Their values are never used: you do not write into your machine the settings of somebody else's machine. Internet is required.

- **Open a shared profile**: Other device page, **"Open a profile..."** → **"Online library..."**. The list gives the brand, model, protocol and number of rows; the "Filter: brand, model, protocol..." field narrows it. **"Open in the table"** (or double-click) loads the profile. Once connected, only profiles of the same protocol can be opened: disconnect to change protocol. Check the names and scales against the device's manual before writing.
- **Share my profile**: in the same window, **"Share my profile..."** and choose one of your profiles. DunenScope makes a copy without values or device number, then opens the sharing form in the browser: attach the copy to it. It will be public; the publisher reviews it before adding it to the library.

22. Safety

By default, DunenScope only reads from the controller. Writing parameters and running motor tests require the full version (section 3) and changes to be allowed, with confirmation (banner of the Settings and Setup wizard pages, or the "Changes allowed (expert)" box on the Parameters page).

- **Automatic backup**: full backup before every write, in Documents\DunenScope\sauvegardes. If the backup fails, the write is cancelled.
- **Read-back** of every written value, compared with the requested value.
- **Dangerous parameters blocked**: CAN address and bit rate, restore defaults. Measurements and factory calibration are read-only. Out-of-range entries are rejected. Stronger confirmation for sensitive parameters.
- **Motor stopped**: required for any write.
- **Motor tests**: STOP or Esc = immediate stop. Automatic stop after 2 minutes of running, when leaving the page, or if the connection or expert mode is lost.
- **First use** of writing and motor tests: motor decoupled or machine made safe, hand on STOP.
- **Other devices**: writing, actions and motor test in the full version (or in simulation) only, with confirmation, automatic backup and read-back. Read-only models, batteries, chargers, J1939 machinery and DBC files receive no setting. The CAN analyser and the listen-only protocols send nothing on the bus. During a test, the motor stops if DunenScope stops responding for one second; with CANopen®, keep the emergency stop within reach.

After a write, check that the values are retained: switch the key off for 10 s, reconnect, read the parameters again and check that the written values are still there.

23. Technical specifications

Item	Value
Operating system	Windows 10 / 11, 64-bit; window usable from 1280 × 680
Dunen protocol	Modbus RTU over CAN, same as DunenConfiger
CAN identifiers	0x602 (PC → controller), 0x6D2 (controller → PC), standard frames
Request size	24 registers maximum
CAN bit rate	250 kbit/s default; 125, 500, 1000 kbit/s
Node	1 by default (1 to 10)
Sampling rate	about 20 samples/s (Dunen)
Controller data	446 parameters, 25 debug variables (Dunen)
Other devices	26 protocols in 4 families: motor controllers, industrial drives, batteries and chargers, machinery and CAN bus (chapter 16)
Links	CAN; serial (RS-485, UART, VE.Direct); USB; Ethernet (Modbus TCP)
CAN adapters	PEAK PCAN-USB, Kvaser, IXXAT, Vector, candleLight (FYSETC UCAN, BTT U2C...), SLCAN (CANable, DunenScope Link...), USB-CAN Analyzer (Seeed, Waveshare); Simulation mode without hardware
Listen without transmitting	PEAK, Kvaser, Vector, candleLight (if it announces it), DunenScope Link

Item	Value
Oscilloscope	4 channels; duration 1 to 60 s; 10, 25 or 50 % before the trigger
Black box	20 s of measurements before each fault; 200 files kept
Languages	French, English, Simplified Chinese, Spanish, German, Italian, Portuguese (Brazil), Russian, Japanese, Korean
Licence	Free discovery mode; full version with a key (lifetime or subscription, one PC per key); signed updates
Exports	DunenConfiger-compatible xlsx; CSV (parameters, measurements, oscilloscope, CAN trace, black box); support package .zip; end-of-line HTML report; commissioning report PDF; profiles (.dsfiche.json)
Component shipped separately	python-can (LGPL-3.0), "bibliotheques" folder

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24. Release status

Version 0.8.0: "Other device" page (controllers compatible with VESC® Software, Kelly, Votol, Fardriver, Bafang and ODrive; CANopen® and Modbus drives; JK, Daly, JBD, Pylontech, Victron, Elcon batteries and chargers; J1939 machinery; DBC file), Overview with the other controllers on the bus, fault log and black box, 4-channel oscilloscope, CAN analyser, PDF commissioning report, online library of profiles, candleLight and USB-CAN Analyzer adapters, python-can shipped separately. These protocols follow the manufacturers' documentation and have been tried with the simulated devices; their validation on real hardware is in progress. Version 0.7.1: point and comma accepted everywhere, window suited to small screens (1280 × 680). Version 0.7.0: "Setup wizard" (motor, controller, command, machine, writing, encoder zero, direction of rotation, PI), "Settings" page by category in the machine's units, mouse-editable curves, winch, conveyor and fan machine types. Version 0.6.0: automatic connection, Other device mode, easy setup, Simple / Workshop / Manufacturer levels, help on every parameter, consistency check before writing, fault log, Help button to write directly to the publisher, error reports. Version 0.4.2: online purchase open (lifetime, yearly, quarterly) and customer portal; updates free for everyone. Version 0.4.0: licensing (discovery mode, online purchase, publisher keys), signed automatic updates, interface and parameter files in 10 languages. Dunen parameter reading and writing used on a real controller (electrified IMER Taurus 1300 mini dumper); validation of motor tests and automatic PI tuning in progress.